Report

Assignment 3, MongoDB: The Movies Dataset

Group: 16
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1 Introduction

In this assignment, we analysed The Movies Dataset from Kaggle. It contains about 45,000 movies with metadata, cast and crew, and plot keywords, plus 26 million ratings from 270,000 MovieLens users. The main tasks were to explore and clean the data, design and populate a MongoDB database, and answer ten questions about it using MongoDB aggregation pipelines and Python. Unlike the Porto taxi data in assignment 2, most of this data is semi-structured: genres, production companies, cast and crew are nested lists stored as text in the CSV files.

Group work

At the beginning of the project, we met in person to go through the assignment and agree on the structure of the repository: shared modules for the database connection and the cleaning rules (DbConnector.py and cleaning.py), and one program for each part of the assignment (see Table 1). Olav carried out the exploratory data analysis (EDA) in part1_eda.py, while Ingrid started on an outline of the solution for the database design and the queries. We went through the outline together and discussed it verbally instead of writing a separate design document. The insert program and the queries (part1_insert.py and part2_queries.py) were then implemented through pair programming, using AI-assisted development. In addition, both of us made smaller fixes and improvements ad hoc throughout the project.

We stayed in contact throughout the project, and discussed questions, changes, and problems both in person and online. When one of us changed or added something that affected the other parts of the project, we reviewed it to make sure the different components remained consistent. Our main communication channel was Messenger. We also used Git for version control, allowing us to keep track of changes and maintain a shared version of the code. The repository is available at: *https://github.com/olavhalleraker/tdt4225-project-2*.

Setup

We ran MongoDB 7.0.1, the version given in the assignment, locally in its own Docker container. Following the database setup in the assignment text, we created an admin user and a user with the `readWrite` role on the database `movies_db`, which the programs use. The programs read the connection settings from environment variables, and the dataset must be extracted to `data/`. We used the full `ratings.csv` (26,024,289 ratings) and `links.csv`, not the small versions. The insert program drops and recreates the two collections, and once the data is loaded, the query program can be rerun independently. The programs use Python 3 with the packages in `requirements.txt` (PyMongo 4.10.1, tabulate, and matplotlib for the EDA figures). We kept `DbConnector.py` from the assignment material, modified to read the settings from environment variables and to fail early if it cannot connect. For more information about the setup, see `README.md`.

File	Purpose
DbConnector.py	Connection to MongoDB
cleaning.py	Parsing and cleaning rules shared by the EDA and the insert program
part1_eda.py	EDA of the raw CSV files, prints tables and saves figures
part1_insert.py	Collections with validators, insertion of the cleaned data, indexes
part2_queries.py	One method per task, with the results printed with pprint and tabulate
explain_indexes.py	explain() of selected queries, used to justify the indexes
output/	Console output of the programs, and CSV files with the complete answers of tasks 2, 5 and 10

Table 1: Overview of files in the repository.

2 Results

The results section follows the three parts of the assignment. First, we explain how we performed the EDA and what we found when analysing and cleaning the dataset. Then, we describe the database design and the insertion of the cleaned data. Finally, we answer the 10 questions using MongoDB aggregation pipelines and Python. The console output of the programs is included as screenshots (Outputs 1–18) of the files in `output/`.

2.1 Exploratory data analysis

The EDA is implemented in `part1_eda.py`. As in assignment 2, it reads the raw CSV files with Python's csv module, and the 26 million ratings are streamed instead of loaded into memory. The nested columns (genres, belongs_to_collection, cast, crew, ...) are not valid JSON but Python literals with single quotes, so they are parsed with `ast.literal_eval`. For each file, the EDA checks the row structure, duplicate IDs, missing and suspicious values and the value distributions. It also checks how well the

files can be joined: credits, keywords and links on the TMDB ID, and ratings through `links.csv`. The statistics are computed with the same functions as the insert program (`cleaning.py`), so the numbers correspond to the data stored in MongoDB. Table 2 summarises the main findings and how each one is handled during insertion. The complete console output is given in Appendix A, with the 17 checks referred to as A1–A17.

Attribute	Finding	Handling
Row structure (movies_metadata)	3 movies are split into a 10-field and a 15-field row by a line break in the overview. Read as they are, the second row has a date as id and production companies as genres (A2).	The two parts are joined again, and all 3 movies are kept.
id (movies_metadata)	45,463 rows but 45,433 unique IDs. 29 IDs occur 2–3 times (30 extra rows). 16 groups are exact copies and the others differ only in popularity (13) and vote_count (2) (A3).	Keep the copy with the highest vote_count (the first one on ties). The TMDB ID becomes _id.
budget, revenue	0 for 80% and 84% of the movies. 244 budgets and 154 revenues are between 1 and 999 USD, e.g. Modern Times with budget 1 and The Karate Kid, Part II with budget 113 (A4, A6, Figure 2).	0 and amounts below 1,000 USD are stored as null. 8,636 budgets and 7,244 revenues are known.
runtime	257 empty and 1,558 zero. 67 runtimes are above 6 hours, mostly TV mini-series (max. Centennial, 1,256 min; A4, A6).	0 and empty are stored as null. Long runtimes are kept.
vote_average, vote_count	2,896 movies have no votes and vote_average 0. 99 more have votes but still average 0 (81 of them one vote), while the lowest real average is 0.5 (A6).	vote_average is null when it is 0. vote_count is kept as it is (0 is a real count).
release_date	84 missing. From 1874 to 2020, 31 movies are dated after the last rating (2017-08-04), and 14 of them are not marked as Released (A8).	Stored as a BSON date (null when missing).
overview, tagline, homepage	959 empty overviews and 142 placeholders such as "No overview found.". 55% have no tagline and 83% no homepage (A4).	Empty strings and placeholders are stored as null.
Nested columns	2,442 movies (5.4%) have no genre. Production companies are empty for 26%, countries for 14% and spoken languages	Parsed, keeping only fields that carry information (genre names, company ID and name, ...). Empty lists are stored as [].

Attribute	Finding	Handling
	for 8%. 4,488 movies (9.9%) belong to one of 1,695 collections, 628 of which have ≥ 3 movies (A9–A11).	
original_language, status, imdb_id	11, 81 and 17 missing (A4). 89 language codes, 71% en (A7).	null when missing.
credits.csv	43 IDs occur more than once (44 extra rows). The copies have the same cast and crew members (7 differ only in the order of the crew list). 1 movie has no credits row, and 2,415 movies have an empty cast and 772 an empty crew (A12, A13).	First copy kept, cast and crew embedded in the movie. Empty lists are stored as [].
Gender (cast, crew)	Code 0 (not set) for 40% of the cast entries and 28.5% of the top-5 billed cast (A13, Figure 6).	Stored as "female", "male" or null.
Cast order	Of the 43,018 movies with a cast, only 31,576 movies have consecutive orders 0..n−1. 11,355 have gaps and 87 duplicate order values (A13).	Kept. Task 6 sorts by order instead of using the list position.
People	3,053 names are used by more than one person ID (e.g. Colin Campbell, 6 IDs). In 951 movies an actor has more than one role (A13).	People are grouped by person ID. An actor counts once per movie.
Directors	888 movies have no director, 3,195 have more than one (up to 42) (A13).	Every director of a movie is counted.
keywords.csv	987 duplicate rows, all exact copies. 14,341 movies have no keywords (A14).	First copy kept, keyword names embedded.
links.csv	219 MovieLens movies have no tmdbId, 161 point to a TMDB ID that is not in the metadata, and 29 TMDB IDs have two MovieLens IDs (A15).	Stored as movielens_ids in the movie and used to link the ratings.
ratings.csv	26,024,289 ratings from 270,896 users, values 0.5–5 in steps of 0.5, 1995–2017. 42,707 ratings (0.16%) cannot be linked to a movie, and in 125 cases a user rated two MovieLens IDs of the same TMDB movie (A16).	All ratings are kept. movie_id is null when the rating cannot be linked. movielens_id is always stored.

Table 2: Findings from the EDA and how they are handled.

```
================================================================================
2. movies_metadata.csv - rows split by a line break in the overview
================================================================================
  Fields in CSV row  Rows
-------------------  ------
                 10  3
                 15  3
                 24  45,460

The 15-field rows read with the 24 column names (all columns are shifted):
id          adult                          genres                                              original_language
----------  -----------------------------  --------------------------------------------------  -------------------
1997-08-20  - Written by Ørnås             [{'name': 'Carousel Productions', 'id': 11176},                    104
2012-09-29  Rune Balot goes to a casino c  [{'name': 'Aniplex', 'id': 2883}, {'name': 'GoHa                    68
2014-01-01  Avalanche Sharks tells the st  [{'name': 'Odyssey Media', 'id': 17161}, {'name'                    82

After joining each 10-field row with the 15-field row that follows it:
    id  title                              release_date    genres
------  ---------------------------------  --------------  ---------------------------------------
 82663  Midnight Man                       1997-08-20      Action, Thriller, Drama
122662  Mardock Scramble: The Third Exhaust  2012-09-29      Animation, Science Fiction
249260  Avalanche Sharks                   2014-01-01      TV Movie, Action, Horror, Science Fiction
-> 3 movies repaired, 45,463 movie rows in total.
```

Output 1: Check A2, rows split by a line break in the overview.

```
================================================================================
3. movies_metadata.csv - duplicate ids
================================================================================
---------------------------------------  ------
Movie rows                               45,463
Unique ids                               45,433
Ids that occur more than once            29
Extra (duplicate) rows                   30
Duplicate groups that are exact copies   16
Groups where popularity differs          13
Groups where vote_count differs          2
---------------------------------------  ------

Examples of copies that differ (kept = highest vote_count, the first one on ties):
id      title                          vote_count    popularity    Decision
------  -----------------------------  ------------  ------------  -----------
132641  Wife                           0             0.096079      kept
132641  Wife                           0             0.619388      dropped
22649   A Farewell to Arms             29            1.914697      kept
22649   A Farewell to Arms             29            2.411191      dropped
84198   A Place at the Table           7             0.501046      kept
84198   A Place at the Table           7             1.673307      dropped
10991   Pokémon: Spell of the Unknown  143           10.264597     dropped
10991   Pokémon: Spell of the Unknown  144           6.480376      kept
-> 45,433 unique movies are inserted.
```

Output 2: Check A3, duplicate movie IDs.

```
================================================================================
4. movies_metadata.csv — missing values per column (after repair and deduplication)
================================================================================
Column                Empty    0 / [] / placeholder    Stored as None or []    Share
--------------------  -------  ----------------------  ----------------------  -------
adult                 0        0                       0                       0.0 %
belongs_to_collection 40,945   0                       40,945                  90.1 %
budget                0        36,553                  36,797                  81.0 %
genres                0        2,442                   2,442                   5.4 %
homepage              37,659   0                       37,659                  82.9 %
id                    0        0                       0                       0.0 %
imdb_id               17       0                       17                      0.0 %
original_language     11       0                       11                      0.0 %
original_title        0        0                       0                       0.0 %
overview              959      142                     1,101                   2.4 %
popularity            0        0                       0                       0.0 %
poster_path           383      0                       –
production_companies  0        11,868                  11,868                  26.1 %
production_countries  0        6,279                   6,279                   13.8 %
release_date          84       0                       84                      0.2 %
revenue               0        38,035                  38,189                  84.1 %
runtime               257      1,558                   1,815                   4.0 %
spoken_languages      0        3,827                   3,827                   8.4 %
status                81       0                       81                      0.2 %
tagline               25,032   0                       25,032                  55.1 %
title                 0        0                       0                       0.0 %
video                 0        0                       0                       0.0 %
vote_average          0        2,995                   2,995                   6.6 %
vote_count            0        2,896                   0 kept
budget/revenue: also None when below 1000 USD. vote_average: None when 0 (no votes or 0 with votes).
poster_path is not stored. Placeholders: No Overview, No overview found., No overview yet.
```

Output 3: Check A4, missing values per column after repair and deduplication.

Missing values

Missing values are coded in many different ways in this dataset: empty strings, 0 (budget, revenue, runtime, vote_average), the text "No overview found.", empty lists `[]`, and gender code 0. We convert all of them to one representation, `null` for single values and `[]` for lists. This matters for the queries, because MongoDB's `$avg`, `$min` and `$max` ignore `null`, while a 0 would silently pull an average or median down. Figure 1 shows how much is missing after cleaning. Revenue and budget are only known for 16% and 19% of the movies, so tasks 1, 4 and 8 rest on a small subset of the movies, which is mostly American and recent: 77% of the movies with known revenue have the US as production country and 79% are from 1990 or later, compared to 47% and 65% of all movies. We report how many movies have a known revenue in those tasks. Nothing is removed apart from the exact or near-duplicate rows. A movie without a revenue still counts for the other questions.

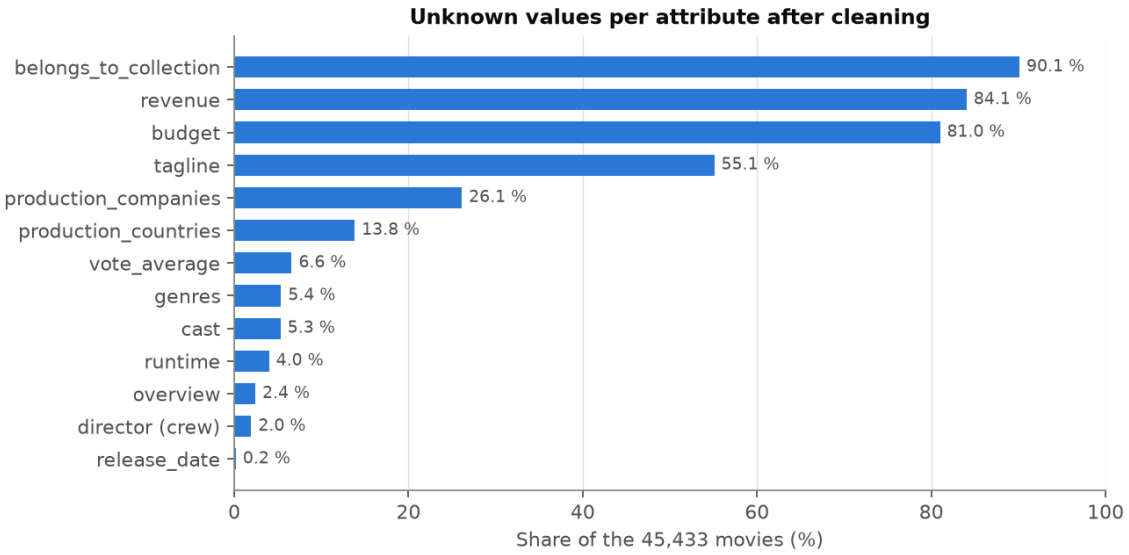


Figure 1: Share of movies where an attribute is unknown after cleaning.

Buggy data

The most serious error is the 3 rows split by a line break (see output 1A2). Read as they are, they create 12 extra "genres" such as *Carousel Productions* and *Aniplex*, which would show up in tasks 3, 5 and 10. Since the 15-field row always follows the 10-field row and the column counts add up exactly (10 + 15 − 1 = 24, because the overview is cut in two), the rows could be repaired instead of dropped.

Figure 2 shows budget against revenue for the movies where both are positive. Most of the movies lie close to the diagonal, mainly between 10^6 and 10^9 USD. The orange points have a budget or revenue below 1,000 USD, which is not realistic for a released film. Many of these look like amounts in millions (*East of Eden*: budget 1, revenue 5). We chose not to multiply them by a million, because some are clearly other mistakes (*Death at a Funeral*: budget 9,000,000, revenue 46), and stored them as unknown instead. The limit of 1,000 USD is a judgement call. It keeps the 54 budgets and 83 revenues between 1,000 and 9,999 USD, which can be real for small films.

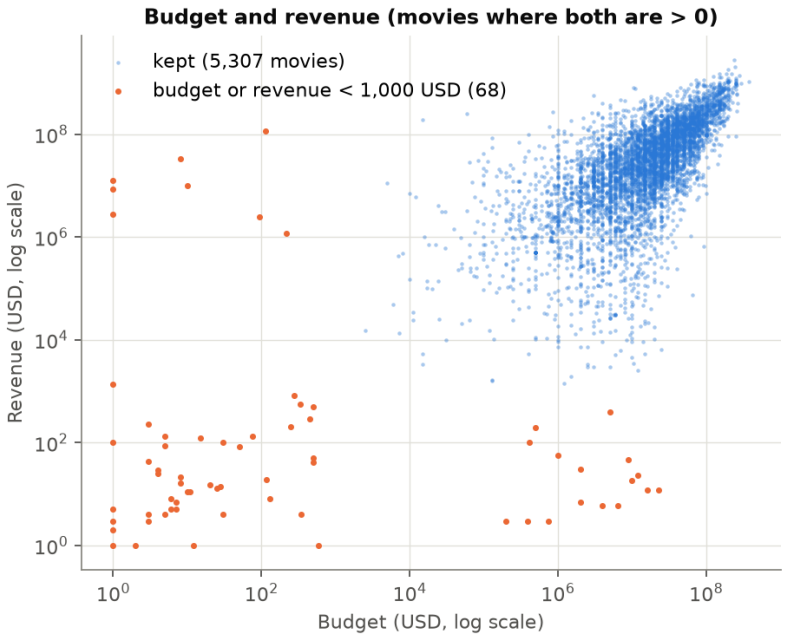


Figure 2: Budget and revenue (log scale) for the movies where both are above 0. Orange points have a budget or revenue below 1,000 USD, which is stored as unknown.

Duplicates are found in every file: movies_metadata, credits and keywords. In all three, the copies either are identical or differ only in values that change over time (popularity, vote count, order of the crew list), so keeping one copy loses no information. Finally, the references between the files are not complete: one movie has no credits, and 0.16% of the ratings point to MovieLens movies without a usable TMDB ID. Because MongoDB has no foreign keys, such dangling references would not be caught by the database, so we resolve them during the insert and store `null` (see 2.2).

```
================================================================================
6. movies_metadata.csv – suspicious values
================================================================================
-------------------------------------  -----
Budget > 0 but below 1,000 USD           244
Revenue > 0 but below 1,000 USD          154
Budget 1,000 to 9,999 USD (kept)          54
Revenue 1,000 to 9,999 USD (kept)         83
vote_average 0 although vote_count > 0    99
vote_average 0 with only 1 vote           81
Lowest vote_average above 0                0.5
Runtime above 6 hours (kept)              67
Movies marked adult                        9
-------------------------------------  -----

Tiny budgets/revenues, the most voted movies (set to None, i.e. unknown):
title                          year    budget    revenue    vote_count
-----------------------------  ------  --------  ---------  ------------
Modern Times                   1936    1         8500000    881
Ong-Bak: The Thai Warrior      2003    0         15         556
Irreversible                   2002    0         792        550
Death at a Funeral             2007    9000000   46         508
The Karate Kid, Part II        1986    113       115103979  457
I Spit on Your Grave 2         2013    0         809        281
2:22                           2017    0         422        277
Dr. Horrible's Sing-Along Blog 2008    200000    3          236

Longest runtimes (mostly TV mini-series, kept):
title                          year    runtime (min)
-----------------------------  ------  ---------------
Centennial                     1978             1256
Jazz                           2001             1140
Baseball                       1994             1140
Berlin Alexanderplatz          1980              931
Heimat: A Chronicle of Germany 1984              925
The Story of Film: An Odyssey  2011              900
```

Output 4: Check A6, suspicious values.

Distributions

Figures 3–7 show the distributions of the main attributes. Drama (45%) and comedy (29%) dominate the genres, and the first genre in the list is usually the broad one. Thriller is the third most common genre, but it is listed first for only 1,663 movies (Figure 3). The number of movies grows strongly after 1990, and 2000–2017 alone holds more than half of the movies (Figure 4). The median runtime is 95 minutes (A5, Figure 5, left). Short films below 30 minutes are common in the earliest decades, which matters for task 5 (see the results of task 5). Movies with few votes have a much wider spread of `vote_average` than movies with ≥ 50 votes, which is why tasks 7 and 8 filter on `vote_count` (Figure 5, right). Women are under-represented in the

top-billed cast, and the share of unknown gender increases further down the cast list (Figure 6). The ratings are skewed towards 3 and 4, and half of the users have rated 30 movies or fewer, so the limit of ≥ 20 ratings in task 10 excludes 39% of the users (Figure 7).

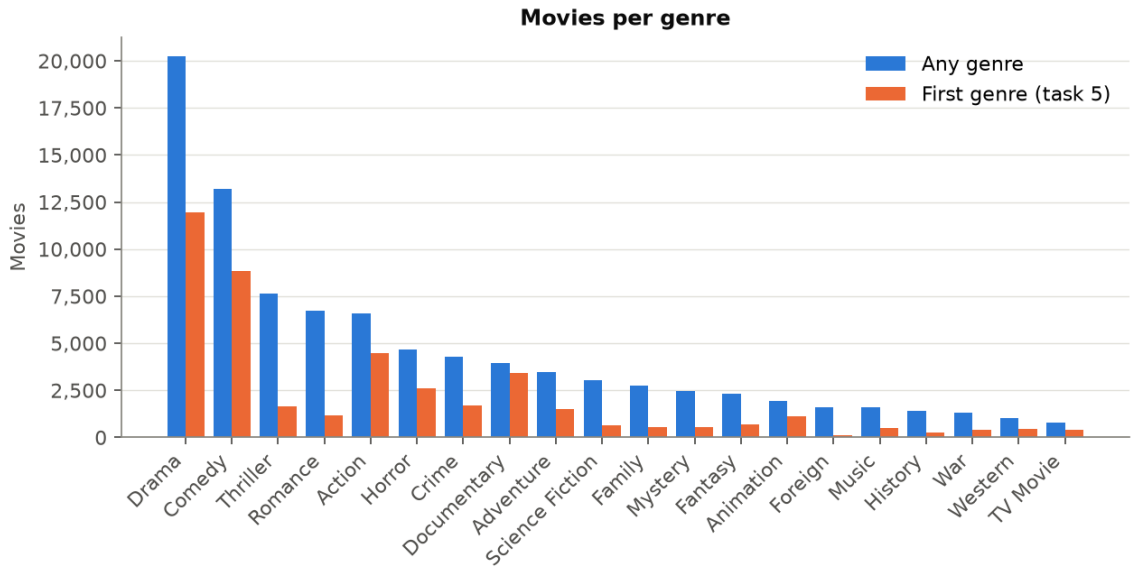


Figure 3: Movies per genre, counting all genres of a movie (blue) and only the first genre (orange).

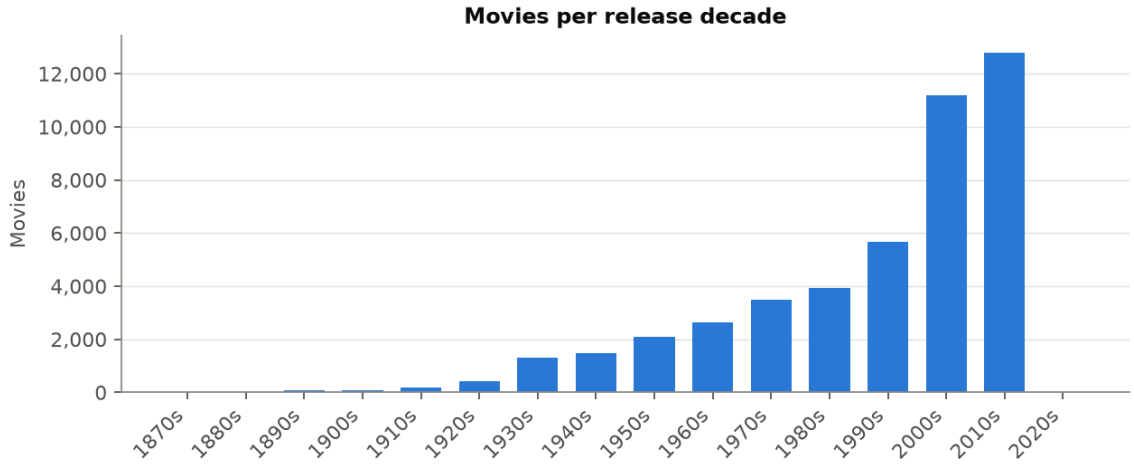


Figure 4: Movies per release decade.

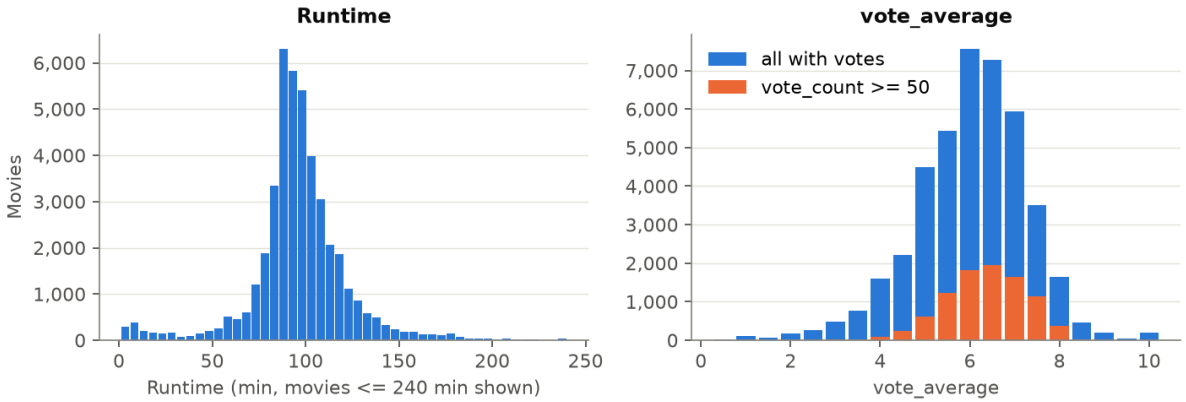


Figure 5: Runtime (left) and vote_average for all movies with votes and for movies with at least 50 votes (right).

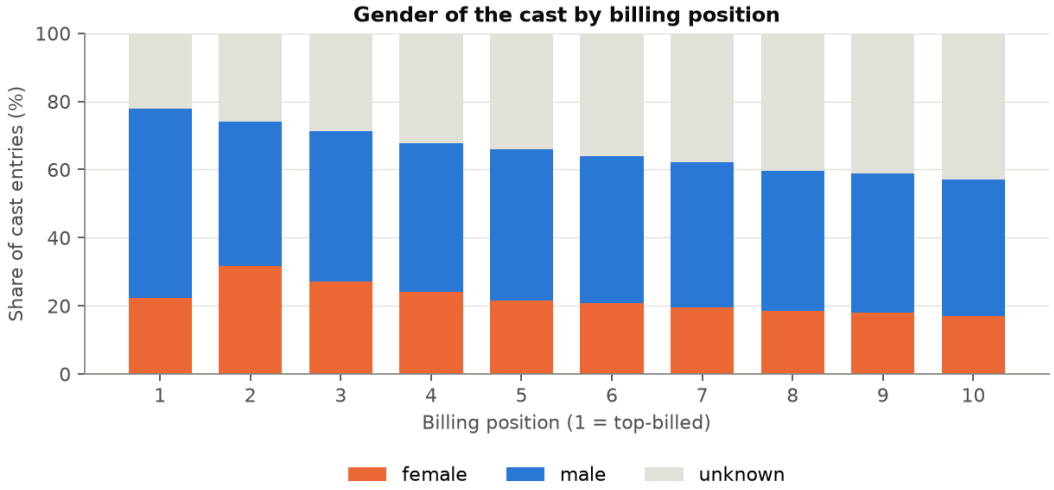


Figure 6: Gender of the cast by billing position (order) for the first ten positions, over all movies.

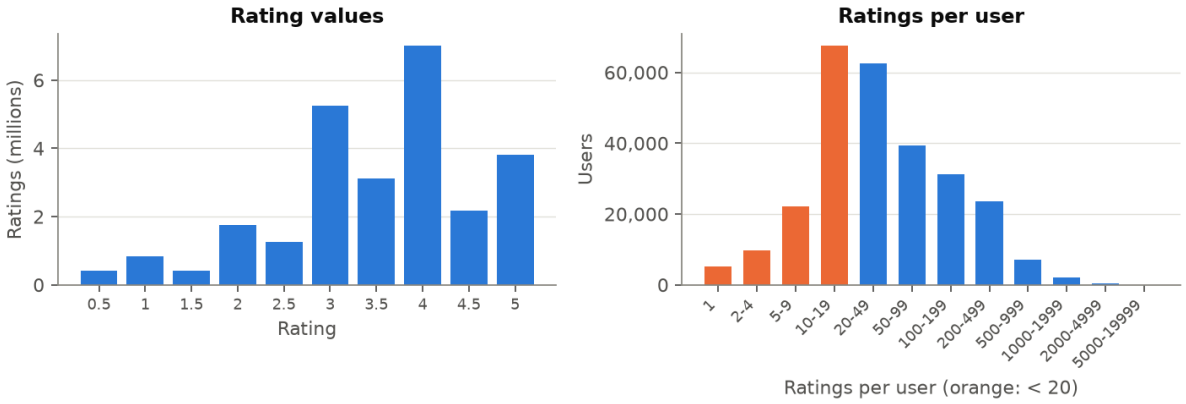


Figure 7: Distribution of the rating values (left) and number of ratings per user (right).

```
================================================================================
17. Impact of the missing values on the Part 2 tasks
================================================================================
Group          Movies    budget known    revenue known    runtime known    vote_average known
-------------  --------  --------------  ---------------  ---------------  --------------------
1870s          2         0.0 %           0.0 %            100.0 %          100.0 %
1880s          4         0.0 %           0.0 %            100.0 %          100.0 %
1890s          75        0.0 %           0.0 %            100.0 %          100.0 %
1900s          87        2.3 %           0.0 %            100.0 %          90.8 %
1910s          176       4.0 %           2.3 %            99.4 %           89.8 %
1920s          431       9.0 %           3.7 %            98.1 %           89.1 %
1930s          1,315     6.0 %           2.9 %            99.0 %           84.9 %
1940s          1,493     5.2 %           3.4 %            98.9 %           91.4 %
1950s          2,079     6.8 %           4.5 %            97.7 %           93.2 %
1960s          2,621     8.6 %           6.1 %            97.3 %           92.5 %
1970s          3,471     10.4 %          7.3 %            96.5 %           90.5 %
1980s          3,928     18.7 %          22.5 %           96.7 %           94.5 %
1990s          5,676     22.5 %          22.9 %           96.1 %           93.2 %
2000s          11,195    25.2 %          19.7 %           95.5 %           94.4 %
2010s          12,795    22.4 %          17.5 %           95.0 %           94.9 %
2020s          1         0.0 %           0.0 %            0.0 %            0.0 %
US production  21,142    29.3 %          26.2 %           99.2 %           95.4 %
Not US         24,291    10.1 %          7.0 %            93.2 %           91.7 %

                          Movies    US production    Released 1990 or later
------------------------  --------  ---------------  ------------------------
Movies with known revenue  7,244     77 %             79 %
All movies                 45,433    47 %             65 %
-> Missing budget/revenue is not spread evenly over decades and countries (compare the rows),
   so medians and totals of revenue (tasks 1, 4 and 8) mostly describe the movies where it is known.

-----------------------------------------------------  -------------
Task 1: directors with >= 5 movies                     2,281
Task 1: ... of these with >= 5 known revenues          382
Task 4: collections with >= 3 movies                   628
Task 4: ... with all revenues known                    129
Task 4: ... with no revenue known                      274
Task 5: movies with date and genre but unknown runtime  1,271
Task 7/8: movies with vote_count >= 50 / >= 100        9,142 / 6,050
-----------------------------------------------------  -------------

Task 6: share of the top-5 billed cast with unknown gender, per decade:
Decade    Top-5 cast entries    Unknown gender
--------  --------------------  ----------------
1870s     2                     50.0 %
1880s     1                     100.0 %
1890s     68                    97.1 %
1900s     96                    77.1 %
1910s     775                   45.0 %
1920s     1,928                 38.6 %
1930s     6,296                 24.5 %
1940s     7,153                 17.6 %
1950s     9,943                 20.6 %
1960s     12,265                29.0 %
1970s     16,079                31.2 %
1980s     18,118                28.1 %
1990s     25,876                21.7 %
2000s     48,659                27.7 %
2010s     54,975                34.0 %
2020s     5                     0.0 %
-> Unknown gender is ignored in task 6. Where the share is high, the female proportion is
   based on fewer people per movie.

Figures written to report/figures/
```

Output 5: Check A17, impact of the missing values on the Part 2 tasks.

2.2 Schema and insertion

The data is stored in two collections: `movies` with one document per movie, and `ratings` with one document per rating. Credits, keywords and MovieLens IDs are embedded in the movie, while a rating refers to its movie by ID. Below is a movie document as returned by `find_one` (with the cast, crew and keywords cut to the first entries), and a rating document:

```
{'_id': 862,
 'imdb_id': 'tt0114709',
 'title': 'Toy Story',
 'original_title': 'Toy Story',
 'original_language': 'en',
 'overview': "Led by Woody, Andy's toys live happily in his room until Andy's ...",
 'tagline': None,
 'status': 'Released',
 'release_date': datetime.datetime(1995, 10, 30, 0, 0),
 'runtime': 81,
 'budget': 30000000,
 'revenue': 373554033,
 'vote_average': 7.7,
 'vote_count': 5415,
 'popularity': 21.946943,
 'adult': False,
 'video': False,
 'homepage': 'http://toystory.disney.com/toy-story',
 'genres': ['Animation', 'Comedy', 'Family'],
 'belongs_to_collection': {'id': 10194, 'name': 'Toy Story Collection'},
 'production_companies': [{'id': 3, 'name': 'Pixar Animation Studios'}],
 'production_countries': [{'iso_3166_1': 'US', 'name': 'United States of America'}],
 'spoken_languages': [{'iso_639_1': 'en', 'name': 'English'}],
 'keywords': ['jealousy', 'toy', 'boy', 'friendship', ...],
 'cast': [{'id': 31, 'name': 'Tom Hanks', 'character': 'Woody (voice)', 'gender': 'male', 'order': 0}, ...],
 'crew': [{'id': 7879, 'name': 'John Lasseter', 'job': 'Director', 'department': 'Directing', 'gender':
'male'}, ...],
 'movielens_ids': [1]}

{'_id': ObjectId('...'),
 'user_id': 1,
 'movie_id': 197,
 'movielens_id': 110,
 'rating': 1.0,
 'rated_at': datetime.datetime(2015, 3, 9, 22, 52, 9)}
```

Both collections are created with a `$jsonSchema` validator (`MOVIE_SCHEMA` and `RATING_SCHEMA` in `part1_insert.py`), which MongoDB checks for every inserted document. The validators require the core fields (such as title, release_date, genres, cast, crew, budget, revenue and vote_average), check their BSON types (e.g. `release_date` must be a date or `null`), and check value ranges such as `rating` between 0.5 and 5, `vote_average` between 0.5 and 10, and `budget`/`revenue` at least 1,000 when known. Two indexes are created after the insert: a text index on `overview` and `tagline` for task 7, and a unique index on (`user_id`, `movielens_id`) in `ratings`.

Reasoning behind the design

- **Embed what belongs to one movie.** Cast, crew, keywords, genres, companies, countries, languages and collection are bounded one-to-many relationships in the terms of the MongoDB schema design articles. A movie has at most 313 cast and

435 crew entries, and the average movie document is 3.3 KB, far below the 16 MB document limit. Six of the ten tasks combine people or genres with movie attributes such as revenue, vote average, runtime and release date, and tasks 4 and 9 use the embedded collection, companies and countries. With everything embedded, each of these tasks is a single aggregation over the movies collection with `$unwind` instead of a join. The data is also only read after loading, so the usual drawback of embedding, that updates must be repeated in many places, does not apply.

- **Reference from the rating to the movie.** Ratings are "one-to-squillions": the most rated movie has 91,921 ratings and the most active user 18,276. Embedding ratings in the movies would make the movie documents large and unbounded, and every movie query would have to read them. A separate collection where each rating refers to its movie (`movie_id` → `movies._id`) keeps the movie documents small, and task 10 joins the two with `$lookup`.
- **The TMDB ID is the `_id`.** It is unique after removing duplicates, and all files except the ratings use it. The ratings use MovieLens IDs, so we translate them with links.csv during the insert and store both: `movie_id` (TMDB, `null` if it cannot be linked) and `movielens_id`. links.csv therefore needs no collection of its own. The MovieLens IDs of each movie are stored in `movielens_ids`, an array because 29 movies have two.
- **No separate collection for people.** Name and gender are repeated in every credit of a person (denormalised). The tasks only need the name for display and group by the person ID, so a people collection would only add a `$lookup` to every people query. Since the same name can belong to several people, the queries group by ID and never by name.
- **Data types.** Dates are BSON dates: `release_date` at 00:00 and `rated_at` converted from Unix time in UTC, so all dates are in the same format. Budget and revenue are integers (64-bit when needed, the largest revenue is 2.8 billion), runtime and vote count integers, rating and vote average doubles, and adult/video booleans. Only fields with information are kept, e.g. the genre name but not the genre ID, and not the poster paths.
- **Missing values are always `null` or `[]`**, never 0, "" or a missing field, so every document has the same fields. The insert program ensures this, and the validators reject invalid values, for example a budget of 0.
- **Indexes only where they help.** The text index is needed for `$text` in task 7 (a collection can have only one). The unique (`user_id`, `movielens_id`) index works like a composite primary key: it guarantees one rating per user and MovieLens movie, and it also serves lookups of one user's ratings (left prefix). All other tasks aggregate over the entire movies collection (45,433 documents) or all of the

ratings, where an index would not help. Unlike assignment 2, we do not store derived values such as year or decade, because the collection is small enough to compute them with `$year` in each query.

We checked the indexes with `explain()` in `explain_indexes.py`. The text search in task 7 uses the text index and examines only 60 documents, and a lookup of one user's ratings uses the unique index and examines 27 documents. A $group over all ratings, like the first stage of task 10, has the plan `GROUP <- COLLSCAN`, so no index would be used there. We therefore did not add more indexes, since each one costs space (the unique index on ratings alone is 286 MB) and insert time. Both indexes are created after the bulk insert, which is faster than updating them for every inserted document.

```
Query                                                  Collection    Plan (winning)                                                                  Docs examined    Keys examined    Returned
-----------------------------------------------------  ------------  ------------------------------------------------------------------------------  ---------------  ---------------  ----------
Task 7: text search for 'noir'                         movies        TEXT_MATCH (overview_tagline_text) <- FETCH <- IXSCAN (overview_tagline_text)             60               60          60
Ratings of one user (left prefix of the unique index)  ratings       FETCH <- IXSCAN (user_movielens_unique)                                                   27               27          27
Ratings of one movie (no index on movie_id)            ratings       COLLSCAN                                                                        26,024,289                0      66,008

Task 10, $group per user on ratings: GROUP <- COLLSCAN

Sizes in MB:
Collection    Index                   MB
------------  ----------------------  -------------
movies        _id_                    0.5
movies        overview_tagline_text   23.2
movies        (data, storage)         149.6, 75.5
ratings       _id_                    286.9
ratings       user_movielens_unique   285.8
ratings       (data, storage)         2628.3, 780.9
```

Output 6: explain() of selected queries and the size of the indexes (output/explain_indexes.txt).

Insertion

`part1_insert.py` drops and recreates the two collections. It reads and cleans the movies, credits, keywords and links with the functions in `cleaning.py`, builds one document per movie, and inserts them in batches of 1,000 with `insert_many` (45,433 movies in 22 s). The ratings are then streamed from the CSV, linked to their movie through links.csv, and inserted in batches of 50,000 with `insert_many(ordered=False)`, which takes about 2.3 minutes for the 26,024,289 ratings. Finally, the indexes are created. The console output shows that all 45,433 movies and 26,024,289 ratings were inserted, matching the EDA. The ratings take 2.6 GB uncompressed but only 781 MB on disk because of WiredTiger's compression.

```
--------------------------------------------

Created collections movies and ratings (with $jsonSchema validators)

Inserted 45,433 movies in 22 s
-----------------------------------------  ----------------------
Movies                                     45,433
Movies with a credits / keywords row       45,432 / 45,432
Movies with non-empty cast / crew / keywords  43,018 / 44,661 / 31,092
Movies with at least one MovieLens id      45,433
Credits rows without a movie (not inserted)   0
Keyword rows without a movie (not inserted)   0
-----------------------------------------  ----------------------
   5,000,000 ratings inserted (26 s)
  10,000,000 ratings inserted (54 s)
  15,000,000 ratings inserted (81 s)
  20,000,000 ratings inserted (103 s)
  25,000,000 ratings inserted (131 s)

Inserted ratings in 2.3 minutes
Step                              Count
--------------------------------  ----------
Ratings inserted                  26,024,289
Linked to a movie                 25,981,582
movie_id null: no tmdbId          13,503
movie_id null: tmdbId not in movies  29,204

Created indexes in 26 s

Documents per collection:
Collection    Documents      Avg. bytes    Data MB    Storage MB  Indexes
------------  -----------  ------------  ---------  ------------  --------------------------
movies        45,433               3292      149.6          75.5  _id_, overview_tagline_text
ratings       26,024,289            100     2628.3         780.9  _id_, user_movielens_unique

db.movies.find_one({'_id': 862}) with cast, crew and keywords cut to the first entries:
{'_id': 862,
 'imdb_id': 'tt0114709',
 'title': 'Toy Story',
 'original_title': 'Toy Story',
 'original_language': 'en',
 'overview': "Led by Woody, Andy's toys live happily in his room until Andy's "
             'birthday brings Buzz Lightyear onto the scene. Afraid of losing '
             "his place in Andy's heart, Woody plots against Buzz. But when "
             'circumstances separate Buzz and Woody from their owner, the duo '
             'eventually learns to put aside their differences.',
 'tagline': None,
 'status': 'Released',
 'release_date': datetime.datetime(1995, 10, 30, 0, 0),
 'runtime': 81,
 'budget': 30000000,
 'revenue': 373554033,
 'vote_average': 7.7,
 'vote_count': 5415,
 'popularity': 21.946943,
 'adult': False,
 'video': False,
 'homepage': 'http://toystory.disney.com/toy-story',
 'genres': ['Animation', 'Comedy', 'Family'],
 'belongs_to_collection': {'id': 10194, 'name': 'Toy Story Collection'},
 'production_companies': [{'id': 3, 'name': 'Pixar Animation Studios'}],
 'production_countries': [{'iso_3166_1': 'US',
                           'name': 'United States of America'}],
 'spoken_languages': [{'iso_639_1': 'en', 'name': 'English'}],
 'keywords': ['jealousy', 'toy', 'boy', 'friendship'],
 'cast': [{'id': 31,
           'name': 'Tom Hanks',
           'character': 'Woody (voice)',
           'gender': 'male',
           'order': 0},
          {'id': 12898,
           'name': 'Tim Allen',
           'character': 'Buzz Lightyear (voice)',
           'gender': 'male',
           'order': 1}],
 'crew': [{'id': 7879,
           'name': 'John Lasseter',
           'job': 'Director',
           'department': 'Directing',
           'gender': 'male'},
          {'id': 12891,
           'name': 'Joss Whedon',
           'job': 'Screenplay',
           'department': 'Writing',
           'gender': 'male'}],
 'movielens_ids': [1]}
```

Output 7: Console output of part1_insert.py (first part).

```
A repaired movie with unknown values (null), db.movies.find_one({'_id': 82663}):
{'_id': 82663,
 'imdb_id': 'tt0113002',
 'title': 'Midnight Man',
 'original_title': 'Midnight Man',
 'original_language': 'en',
 'tagline': None,
 'status': 'Released',
 'release_date': datetime.datetime(1997, 8, 20, 0, 0),
 'runtime': 104,
 'budget': None,
 'revenue': None,
 'vote_average': 6.0,
 'vote_count': 1,
 'popularity': 0.065736,
 'adult': False,
 'video': False,
 'homepage': None,
 'genres': ['Action', 'Thriller', 'Drama'],
 'belongs_to_collection': None,
 'production_companies': [{'id': 11176, 'name': 'Carousel Productions'},
                          {'id': 11602, 'name': 'Vision View Entertainment'},
                          {'id': 29812,
                           'name': 'Telescene Film Group Productions'}],
 'production_countries': [{'iso_3166_1': 'CA', 'name': 'Canada'},
                          {'iso_3166_1': 'LU', 'name': 'Luxembourg'},
                          {'iso_3166_1': 'GB', 'name': 'United Kingdom'},
                          {'iso_3166_1': 'US',
                           'name': 'United States of America'}],
 'spoken_languages': [{'iso_639_1': 'en', 'name': 'English'}],
 'keywords': ['undercover', 'soldier'],
 'cast': [{'id': 2879,
           'name': 'Rob Lowe',
           'character': 'Sean Dillon',
           'gender': 'male',
           'order': 0}],
 'crew': [{'id': 43673,
           'name': 'Tom Reeve',
           'job': 'Supervisor of Production Resources',
           'department': 'Crew',
           'gender': 'male'}],
 'movielens_ids': [97919]}

db.ratings.find_one({'user_id': 1}):
{'_id': ObjectId('6ae221c2765ff0288cf3c18c'),
 'user_id': 1,
 'movie_id': 197,
 'movielens_id': 110,
 'rating': 1.0,
 'rated_at': datetime.datetime(2015, 3, 9, 22, 52, 9)}

------------------------------------------------
Connection to movies_db-db is closed
```

Output 8: Console output of part1_insert.py (second part).

2.3 Queries

In this part, we make the following assumptions. Unknown values are `null` (see 2.1) and are left out of means and medians: MongoDB's `$avg` ignores `null`, and our median expression filters it out. People are identified by their TMDB person ID, and an actor or director counts once per movie even if listed several times. "Movies" means the movies in our collection. Every task is a method in `part2_queries.py`. It runs one aggregation pipeline (in a few tasks also a `find` or `count_documents` for extra information), prints the first result document as MongoDB returns it with pprint, and then all results as a table with tabulate. Long answers (tasks 2, 5 and 10) are also written in full to CSV files in `output/`.

Three tasks need a median. MongoDB 7.0 has a `$median` accumulator, but only with the method `"approximate"`, which for an even number of values returns one of them instead of the mean of the middle two (we tested that it returns 2 for [1, 2, 3, 4], and 10 for [10, 20]). We therefore build an exact median with the Python function below, which returns an aggregation expression. It sorts the non-null values with `$sortArray` and averages the two middle elements (the same element twice when the count is odd):

```
def median(array):
    values = {"$sortArray": {"input": {"$filter": {"input": array, "cond": {"$ne": ["$$this", None]}}},
                             "sortBy": 1}}
    return {"$let": {
        "vars": {"v": values},
        "in": {"$avg": [
            {"$arrayElemAt": ["$$v", {"$floor": {"$divide": [{"$subtract": [{"$size": "$$v"}, 1]}, 2]}}]},
            {"$arrayElemAt": ["$$v", {"$floor": {"$divide": [{"$size": "$$v"}, 2]}}]},
        ]},
    }}
```

Task 1: Directors with the highest median revenue

The pipeline unwinds the crew, keeps the entries with job Director, and groups first by (director, movie), so that a director listed twice for the same movie counts once, and then by director. Revenue is only known for 16% of the movies, so many of the 2,281 directors with ≥ 5 movies have few known revenues: 974 have none and 617 only one or two, and a median of one or two values says little. We therefore show two lists, and report both the total movie count and the number with known revenue:

1. **Literal reading:** ≥ 5 directed movies, with the median over their known revenues. Joss Whedon tops this list (1.41 billion USD), followed by Kyle Balda (1.02 billion) and David Yates (936 million), but the first two are based on only 3 known revenues each, and Mike Thurmeier and Raman Hui reach the top 10 on only 2.
2. **Stricter reading:** ≥ 5 movies *with known revenue* (382 directors), so that every median is based on at least five values. David Yates, who directed the last four Harry Potter films, has the highest median revenue (936 million USD), followed by

Peter Jackson and George Lucas. Eric Darnell and Tom McGrath have the same median because they co-directed the three *Madagascar* films, and for both the middle value is the revenue of *Madagascar*.

Only 5 directors appear in both lists, which shows how sensitive the answer is to the missing revenues. Both lists are dominated by directors of franchises and animated films (Harry Potter, Lord of the Rings, Star Wars, Shrek, Ice Age).

```
================================================================================
Task 1: Considering only crew with job = Director, which 10 directors with >=
5 movies have the highest median revenue? Also report each director's movie
count and mean vote_average.
================================================================================
Interpretation: directors with >= 5 movies (the median is taken over their known revenues;
movie count and mean vote_average are over all their movies).

First result document:
{'_id': 12891,
 'director': 'Joss Whedon',
 'movies': 5,
 'movies_with_revenue': 3,
 'mean_vote_average': 7.4,
 'median_revenue': 1405403694.0}
```

rank	director_id	director	movies	movies_with_revenue	median_revenue	mean_vote_average
1	12891	Joss Whedon	5	3	1,405,403,694	7.40
2	8023	Kyle Balda	5	3	1,020,063,384	6.62
3	11343	David Yates	10	6	936,085,968	6.70
4	108	Peter Jackson	14	11	871,368,364	7.12
5	1	George Lucas	7	6	712,398,168	6.81
6	124748	Chris Renaud	5	4	709,485,961	6.70
7	229962	Mike Thurmeier	6	2	642,911,910	6.67
8	7087	Brad Bird	6	5	623,722,818	7.15
9	10943	Francis Lawrence	6	6	619,388,636	6.80
10	12099	Raman Hui	5	2	592,121,491	6.22

```
Stricter reading: directors with >= 5 movies WITH KNOWN REVENUE (median over >= 5 values):
```

rank	director_id	director	movies	movies_with_revenue	median_revenue	mean_vote_average
1	11343	David Yates	10	6	936,085,968	6.70
2	108	Peter Jackson	14	11	871,368,364	7.12
3	1	George Lucas	7	6	712,398,168	6.81
4	7087	Brad Bird	6	5	623,722,818	7.15
5	10943	Francis Lawrence	6	6	619,388,636	6.80
6	18863	Eric Darnell	5	5	532,680,671	6.34
7	18864	Tom McGrath	5	5	532,680,671	6.40
8	5714	Carlos Saldanha	7	5	500,188,435	6.04
9	5524	Andrew Adamson	6	5	484,409,218	6.52
10	7879	John Lasseter	10	5	461,983,149	6.72

```
Directors with >= 5 movies: 2,281, of these with >= 5 known revenues: 382
Directors with >= 5 movies by number of known revenues (those with 0 have no median):
known_revenues      directors
----------------  -----------
0                         974
1–2                       617
3–4                       308
>= 5                      382
[task 1 took 8.5 s]
```

Output 9: Result of task 1.

Task 2: Actor pairs with at least 3 movies together

The distinct actors of each movie are paired by unwinding the actor list twice and keeping a pair only when the first person ID is smaller than the second, so each

unordered pair is counted once per movie. This gives about 6.5 million pairs (A13), which are grouped by the two IDs, with `allowDiskUse` enabled. 45,650 pairs have co-starred in at least 3 movies. Huntz Hall and Leo Gorcey from the *Bowery Boys* series lead with 35 movies together, followed by Charlie Chaplin and Edna Purviance (33) and Laurel and Hardy (31). Much of the top of the list consists of comedy duos from film series, actors from long series of TV films and voice actors in animated films (Jeff Bennett, Rob Paulsen, Frank Welker). We count the whole cast and not only the leading roles. Pairs with the same number of movies are ordered by the average vote_average of their shared movies. The table shows the first 30 pairs, and the full list is in `output/task2_actor_pairs.csv`.

```
================================================================================
Task 2: List actor pairs that have co-starred in >= 3 movies together, their
number of co-appearances, and their average movie vote_average. Sort by number
of co-appearances.
================================================================================
Actor pairs with >= 3 movies together: 45,650

First result document:
{'_id': {'actor_1_id': 33024, 'actor_2_id': 89989},
 'actor_1': 'Huntz Hall',
 'actor_2': 'Leo Gorcey',
 'co_appearances': 35,
 'avg_vote_average': 6.3549999999999995}

Top 30 by co-appearances (ties by average vote_average):
  rank  actor_1_id    actor_1           actor_2_id    actor_2           co_appearances    avg_vote_average
------  ------------  ----------------  ------------  ----------------  ----------------  ------------------
     1  33024         Huntz Hall        89989         Leo Gorcey                    35                6.35
     2  13848         Charlie Chaplin   21301         Edna Purviance                33                6.46
     3  78057         Oliver Hardy      89670         Stan Laurel                   31                6.35
     4  29642         Lou Costello      30111         Bud Abbott                    27                6.47
     5  34982         Jeff Bennett      43125         Rob Paulsen                   27                6.23
     6  15761         Grey Griffin      15831         Frank Welker                  25                6.71
     7  7685          Raymond Burr      109701        Barbara Hale                  25                5.93
     8  4165          John Wayne        8496          Paul Fix                      24                5.85
     9  12077         Jim Cummings      15831         Frank Welker                  21                6.55
    10  133277        Frank Mayo        980330        Jack Mower                    21                6.28
    11  12077         Jim Cummings      34982         Jeff Bennett                  21                6.27
    12  15831         Frank Welker      34982         Jeff Bennett                  21                6.23
    13  5             Peter Cushing     113           Christopher Lee               21                5.99
    14  14034         Bernard Gorcey    89989         Leo Gorcey                    20                7.00
    15  13848         Charlie Chaplin   14438         Henry Bergman                 20                6.98
    16  7450          Toshirō Mifune    7453          Takashi Shimura               19                7.48
    17  14034         Bernard Gorcey    33024         Huntz Hall                    19                7.11
    18  8635          Buster Keaton     10525         Joe Roberts                   19                6.88
    19  62410         Sammo Hung        62414         Yuen Biao                     19                6.83
    20  18897         Jackie Chan       62414         Yuen Biao                     19                6.80
    21  13848         Charlie Chaplin   21310         John Rand                     19                6.75
    22  20519         Simon Yam         25251         Lam Suet                      19                6.10
    23  68527         William R. Moses  109701        Barbara Hale                  19                6.05
    24  13848         Charlie Chaplin   21306         Albert Austin                 18                6.74
    25  40941         Kenneth Williams  40944         Charles Hawtrey               18                6.39
    26  19292         Adam Sandler      20818         Allen Covert                  18                5.64
    27  121323        Bess Flowers      171111        Harold Miller                 17                6.96
    28  383           Michael Palin     8930          John Cleese                   17                6.85
    29  18897         Jackie Chan       62410         Sammo Hung                    17                6.84
    30  15831         Frank Welker      86314         Mindy Cohn                    17                6.66
(all 45,650 rows written to output/task2_actor_pairs.csv)
[task 2 took 44.4 s]
```

Output 10: Result of task 2.

Task 3: Actors with the widest genre breadth

The cast entries are grouped per actor with `$addToSet` of {movie ID, genres}, so an actor counts once per movie, and the genres are combined with `$setUnion`. 10,405 actors have at least 10 credited movies. The breadth has a natural maximum: there are only 20 genres, and 22 actors have appeared in all of them (and 60 in 19). The top 10 is therefore decided by the tie-break. We chose the number of credited movies, fewest first: an actor who covers all genres in 43 movies (Eddie Izzard) has a wider range relative to the number of films than one who needed 65 (Michael Madsen). As example genres we show each actor's five most common genres with the number of movies, since the first five of all 20 genres would be the same for every actor in the list.

```
======================================================================
Task 3: List the top 10 actors (with >= 10 credited movies) that have the
widest genre breadth. Report the actor, the number of distinct genres they've
appeared in, and up to 5 example genres.
======================================================================
Actors with >= 10 credited movies: 10,405. Of these, 22 have all 20 genres and 60 have 19.

First result document:
{'_id': 1926,
 'actor': 'Eddie Izzard',
 'credited_movies': 43,
 'distinct_genres': 20,
 'example_genres': [{'genre': 'Comedy', 'movies': 19},
                    {'genre': 'Drama', 'movies': 15},
                    {'genre': 'Thriller', 'movies': 9},
                    {'genre': 'Family', 'movies': 7},
                    {'genre': 'Adventure', 'movies': 6}]}

Top 10 by distinct genres (ties: fewer credited movies first). Example genres = the actor's
5 most common genres, with the number of the actor's movies in that genre:
  rank  actor_id    actor             credited_movies    distinct_genres  example_genres
------  ----------  ---------------  -----------------  -----------------  ------------------------------------------------------------------
     1  1926        Eddie Izzard                    43                 20  Comedy (19), Drama (15), Thriller (9), Family (7), Adventure (6)
     2  11998       Eddie Albert                    50                 20  Drama (25), Comedy (17), Romance (11), Action (7), Adventure (6)
     3  378         Jonathan Pryce                  55                 20  Drama (25), Comedy (15), Action (13), Adventure (11), Thriller (11)
     4  825         Stacy Keach                     59                 20  Drama (28), Action (13), Comedy (13), Thriller (13), Adventure (10)
     5  9191        Timothy Spall                   59                 20  Drama (37), Comedy (15), Fantasy (13), Family (12), Adventure (11)
     6  2222        Beau Bridges                    60                 20  Drama (38), Comedy (19), Action (13), Romance (11), Adventure (9)
     7  141         David Carradine                 60                 20  Action (28), Drama (21), Thriller (17), Comedy (16), Adventure (10)
     8  2440        Bill Nighy                      61                 20  Comedy (29), Drama (29), Action (16), Family (13), Adventure (12)
     9  1954        Billy Zane                      64                 20  Drama (31), Thriller (24), Action (21), Comedy (18), Horror (12)
    10  147         Michael Madsen                  65                 20  Thriller (31), Drama (27), Action (26), Crime (14), Horror (12)
[task 3 took 2.0 s]
```

Output 11: Result of task 3.

Task 4: Collections with the largest total revenue

Movies that belong to a collection are grouped by the collection ID. Collections with fewer than 3 movies are removed, and the rest are sorted by the sum of the revenue. *Harry Potter* (7.7 billion USD from 8 films) narrowly beats *Star Wars* (7.4 billion) and *James Bond* (7.1 billion from 26 films). Unknown revenue is left out of the sum, so the total is a lower bound when `movies_with_revenue` is smaller than `movies`, as for *Despicable Me* where 2 of the 6 revenues are unknown. The release dates show how long each series has run, from 1962 to 2015 for James Bond.

```
================================================================================
Task 4: For film collections (belongs_to_collection.name not null) with >= 3
movies, which 10 collections have the largest total revenue? For each, report
movie count, total revenue, median vote_average, and the earliest -> latest
release date.
================================================================================
First result document:
{'_id': 1241,
 'collection': 'Harry Potter Collection',
 'movies': 8,
 'movies_with_revenue': 8,
 'total_revenue': 7707367425,
 'earliest_release': datetime.datetime(2001, 11, 16, 0, 0),
 'latest_release': datetime.datetime(2011, 7, 7, 0, 0),
 'median_vote_average': 7.5}

  rank  collection                              movies    movies_with_revenue    total_revenue    median_vote_average  earliest -> latest release
------  ------------------------------------  --------  ---------------------  ---------------  ---------------------  ----------------------------
     1  Harry Potter Collection                    8                      8    7,707,367,425                   7.50  2001-11-16 -> 2011-07-07
     2  Star Wars Collection                       8                      8    7,434,494,790                   7.45  1977-05-25 -> 2016-12-14
     3  James Bond Collection                     26                     26    7,106,970,239                   6.30  1962-10-04 -> 2015-10-26
     4  The Fast and the Furious Collection        8                      8    5,125,098,793                   6.65  2001-06-22 -> 2017-04-12
     5  Pirates of the Caribbean Collection        5                      5    4,521,576,826                   6.90  2003-07-09 -> 2017-05-23
     6  Transformers Collection                    5                      5    4,366,101,244                   6.10  2007-06-27 -> 2017-06-21
     7  Despicable Me Collection                   6                      4    3,691,070,216                   6.90  2010-07-08 -> 2017-06-15
     8  The Twilight Collection                    5                      5    3,342,107,290                   5.80  2008-11-20 -> 2012-11-13
     9  Ice Age Collection                         5                      5    3,216,708,553                   6.50  2002-03-10 -> 2016-06-23
    10  Jurassic Park Collection                   4                      4    3,031,484,143                   6.35  1993-06-11 -> 2015-06-09

total_revenue is the sum of the known revenues only (see movies_with_revenue), so it is a
lower bound for collections where some revenues are unknown.
[task 4 took 0.0 s]
```

Output 12: Result of task 4.

Task 5: Median runtime by decade and primary genre

Movies with a release date and at least one genre (42,965) are grouped by decade, computed from `release_date` with `$year`, and the first genre (`$arrayElemAt`). The median uses the 41,694 of them with a known runtime. The answer has 232 rows. The output below shows the first part (the 1870s–1910s), and the complete list is in Appendix B and in `output/task5_runtime_by_decade_and_genre.csv`. Before 1910 almost all movies are short films of 1–15 minutes. From the 1940s on, the medians of most genres lie between 85 and 105 minutes. War and history films are the longest from the 1960s on (often 105–120 minutes), while animation is the shortest. Animation also shows the history of the format: 6–8 minutes from the 1920s to the 1950s (cartoon shorts), 25 minutes in the 1960s, 38 minutes in the 1970s and 70–80 minutes from the 1980s, when feature films take over. Many groups in the early decades contain only a few movies, so their medians are not reliable. For instance, 1940s science fiction (220 minutes) is a single movie. The 2020s group has one movie with unknown runtime, so its median is `null`.

```
================================================================================
Task 5: By decade and primary genre (first element in genres), what is the
median runtime and movie count? List results sorted by decade then median
runtime (desc).
================================================================================
Movies with release date and at least one genre: 42,965 (41,694 with known runtime), in 232 (decade, genre) groups.
Sorted by decade (ascending), then median runtime (descending). Groups with only a few movies
(e.g. before 1910) give medians that say little.

First result document:
{'decade': 1870,
 'primary_genre': 'Documentary',
 'movies': 2,
 'movies_with_runtime': 2,
 'median_runtime': 1.0}
```

decade	primary_genre	movies	movies_with_runtime	median_runtime
1870s	Documentary	2	2	1.00
1880s	Documentary	4	4	1.00
1890s	Fantasy	8	8	1.50
1890s	Documentary	27	27	1.00
1890s	Comedy	9	9	1.00
1890s	Horror	3	3	1.00
1890s	Action	2	2	1.00
1890s	Drama	2	2	1.00
1890s	Family	1	1	1.00
1890s	History	1	1	1.00
1900s	History	1	1	60.00
1900s	Adventure	4	4	15.00
1900s	Drama	7	7	10.00
1900s	Action	2	2	9.00
1900s	Thriller	1	1	8.00
1900s	Science Fiction	2	2	5.50
1900s	Fantasy	15	15	4.00
1900s	Horror	2	2	4.00
1900s	Comedy	17	17	3.00
1900s	Music	1	1	3.00
1900s	Crime	2	2	2.00
1900s	Animation	1	1	2.00
1900s	Documentary	12	12	1.00
1900s	Romance	2	2	1.00
1910s	History	2	2	109.00
1910s	War	2	2	108.00
1910s	Mystery	3	3	84.00
1910s	Adventure	8	8	70.00
1910s	Romance	6	6	68.50
1910s	Drama	52	52	65.00
1910s	Western	4	4	60.50
1910s	Crime	7	7	59.00
1910s	Family	1	1	52.00
1910s	Documentary	2	2	43.50
1910s	Horror	2	2	38.50
1910s	Action	1	1	29.00
1910s	Comedy	54	53	27.00
1910s	Fantasy	6	6	14.50
1910s	Animation	5	5	12.00

Output 13: Result of task 5 (first part, the 1870s–1910s).

Task 6: Female proportion of the top-billed cast by decade

For every movie with a cast and a release date, the cast is sorted by `order` and the first five are kept (`$sortArray` and `$slice`). Cast members with unknown gender are left out, so the proportion is female / (female + male) among the known genders in the top 5. Movies where none of the five has a known gender (3,320 of the 43,000 movies with a cast and a release date) are not used, which leaves 39,680 movies. The proportions of the movies are then averaged per decade, so every movie counts equally. The 1870s, 2020s and 1890s top the list, but they contain only 1, 1 and 2

movies. Among the decades with more than 100 movies, the 2010s have the highest average female proportion (37.6%) and the 1970s the lowest (31.1%). The differences are small, and only roughly form a U shape: about 36% in the 1930s–40s, 31–32% in the 1970s–80s, and the highest value in the 2010s. Since gender is unknown for 28.5% of the top-5 cast entries, and probably not at random, the numbers should be read as approximate. Unknown gender is also most common in the 2010s (34% of the top-5 entries, see A17), so the highest value rests on the least complete data.

```
================================================================================
Task 6: For each movie's top-billed 5 cast (by order), what is the proportion
of female cast? Aggregate by decade and list decades sorted by average female
proportion (desc), including movie counts used. Unknown gender may be ignored.
================================================================================
First result document:
{'_id': 1870, 'avg_female_proportion': 1.0, 'movies': 1}

decade      avg_female_proportion    movies
--------  ----------------------  --------
1870s                      1.0000         1
2020s                      0.8000         1
1890s                      0.5000         2
1900s                      0.4444        12
2010s                      0.3763    10,543
1930s                      0.3655     1,249
2000s                      0.3616     9,647
1940s                      0.3589     1,435
1990s                      0.3375     5,179
1910s                      0.3369       144
1950s                      0.3365     1,998
1920s                      0.3313       374
1960s                      0.3251     2,395
1980s                      0.3200     3,555
1970s                      0.3111     3,145

Movies with a cast and a release date: 43,000, of these with no known gender in the top 5: 3,320
Movies used in total: 39,680
avg_female_proportion is the mean of the per-movie proportions (female / known gender among
the top 5). Decades with only a few movies (e.g. the 1870s, 1890s and 2020s) are not reliable.
[task 6 took 0.3 s]
```

Output 14: Result of task 6.

Task 7: Noir movies with the highest vote_average

The task uses the text index on `overview` and `tagline` with `$text`. MongoDB's text search splits words at hyphens, so "neo-noir" is indexed as the two words "neo" and "noir". Searching for "noir" therefore finds both noir and neo-noir. Adding "neo-noir" to the search string would also match "neo" on its own and add 43 wrong movies, such as *The Matrix Reloaded* (Neo) and *Akira* (Neo-Tokyo). 60 movies match, but only 5 of them have at least 50 votes, so the answer has 5 movies instead of 20: *The Bad Sleep Well*, *Drunken Angel*, *Elevator to the Gallows*, *Synchronicity* and *Frank & Lola*. The text search matches whole words. A regular expression `that looks for "noir" in the overview or tagline matches 67 movies, 7 more than the text search, because it also matches longer words that start with "noir", while the`

`text search only matches the whole word`. With ≥ 50 votes, the regular expression gives only 6 movies (the extra one is Phoenix), so the small answer comes from the data and not from the index. The EDA showed that 398 movies have the keyword "film noir" or "neo-noir" but only 18 of them mention noir in the overview or tagline (A14), so most noir films cannot be found with this search.

```
======================================================================
Task 7: Using a text search over overview and tagline, which 20 movies
matching "noir" or "neo-noir" (and vote_count >= 50) have the highest
vote_average? Return title, year, vote_average, and vote_count.
======================================================================
Text search "noir": 60 movies, 5 of them with vote_count >= 50.
(Searching "noir neo-noir" would add 43 movies where only "neo" matches, e.g. The Matrix Reloaded, The Matrix Revolutions, Akira.)
Cross-check with a regex instead of the text index: 67 movies contain "noir" in overview or tagline, 6 of them with vote_count >= 50.
Found by the regex but not by the text search (vote_count >= 50): Phoenix (147 votes)
So fewer than 20 movies can be returned: few overviews/taglines use the word.

First result document:
{'title': 'The Bad Sleep Well',
 'vote_average': 7.7,
 'vote_count': 57,
 'year': 1960,
 'matched': 'noir'}

  rank  title                   year      vote_average    vote_count  matched
------  ----------------------  ------  --------------  ------------  ---------
     1  The Bad Sleep Well      1960              7.70            57  noir
     2  Drunken Angel           1948              7.70            54  noir
     3  Elevator to the Gallows 1958              7.60            85  noir
     4  Synchronicity           2015              5.70           114  noir
     5  Frank & Lola            2016              5.70            51  noir
[task 7 took 0.2 s]
```

Output 15: Result of task 7.

Task 8: Director-actor pairs with the highest mean vote_average

For each movie with at least 100 votes, the distinct directors and actors are combined by unwinding both lists. Pairs where the director is also the actor are removed, since a director acting in their own film is not a collaboration. As a check, the program shows that none of these self-pairs would have reached the top 20 anyway. Pairs with at least 3 movies are sorted by the mean `vote_average`, then by the number of films and the name. Akira Kurosawa and the actors he worked with fill most of the list (15 of the top 20). Charlie Chaplin with Hank Mann, Francis Ford Coppola with John Cazale and Robert Duvall (*The Godfather* films, among others), Quentin Tarantino with Harvey Keitel, and Hayao Miyazaki with Tsunehiko Kamijô are the other pairs. Many pairs have the same mean because they share the same 3 movies, so their order is decided by the tie-break. At the cut-off, 27 pairs have a mean of at least 8.033, so 7 pairs with the same mean as ranks 17–20 are left out only because of the tie-break. The mean revenue only uses the films with known revenue, so for Kurosawa it is based on one to three films and is very low, and for Susumu Fujita no revenue is known.

```
================================================================================
Task 8: Which 20 director–actor pairs with >= 3 collaborations (same movie)
have the highest mean vote_average, considering only movies with vote_count >=
100? Include the pair's films count and mean revenue.
================================================================================
First result document:
{'_id': {'director_id': 5026, 'actor_id': 106165},
 'director': 'Akira Kurosawa',
 'actor': 'Eijirô Tôno',
 'films': 3,
 'mean_vote_average': 8.133333,
 'films_with_revenue': 1,
 'mean_revenue': 271841.0}
```

rank	director	actor	films	mean_vote_average	films_with_revenue	mean_revenue
1	Akira Kurosawa	Eijirô Tôno	3	8.133	1	271,841
2	Akira Kurosawa	Haruo Suzuki	3	8.133	1	271,841
3	Akira Kurosawa	Minoru Itô	3	8.133	1	271,841
4	Charlie Chaplin	Hank Mann	3	8.133	3	6,506,394
5	Akira Kurosawa	Atsushi Watanabe	3	8.100	2	163,540
6	Akira Kurosawa	Toranosuke Ogawa	3	8.100	2	163,540
7	Francis Ford Coppola	John Cazale	3	8.100	3	99,009,751
8	Quentin Tarantino	Harvey Keitel	3	8.100	3	182,573,606
9	Akira Kurosawa	Daisuke Katô	4	8.075	3	141,216
10	Akira Kurosawa	Ichirô Chiba	4	8.075	2	163,540
11	Akira Kurosawa	Noriko Honma	4	8.075	3	141,216
12	Francis Ford Coppola	Robert Duvall	4	8.075	4	96,622,408
13	Akira Kurosawa	Haruya Sakamoto	3	8.067	1	271,841
14	Akira Kurosawa	Shôichi Hirose	3	8.067	1	271,841
15	Akira Kurosawa	Susumu Fujita	3	8.067	0	
16	Akira Kurosawa	Isao Kimura	4	8.050	2	163,540
17	Akira Kurosawa	Minoru Chiaki	6	8.033	3	141,216
18	Akira Kurosawa	Masao Shimizu	3	8.033	1	55,240
19	Akira Kurosawa	Yûnosuke Itô	3	8.033	1	55,240
20	Hayao Miyazaki	Tsunehiko Kamijô	3	8.033	2	217,150,202

```
mean_revenue is the mean of the known revenues (films_with_revenue).
Pairs with a mean vote_average at least as high as rank 20 (8.033): 27 (more than 20 means that pairs tied at the cut-off are left out by the tie-break)

Directors who act in >= 3 of their own films (excluded above) with a mean vote_average at
least as high as rank 20 (8.03): 0
[task 8 took 1.2 s]
```

Output 16: Result of task 8.

Task 9: Original languages of non-English American movies

The movies with an `original_language` other than `en` (and not missing) are kept if a production country is the US, or if a production company has "United States" in its name. In the data, a production company only has a name and an ID, not a country, so the name is all we can check. The company condition adds no movies beyond the country condition (only *United States Pictures* and *United States Navy* match, see A11). 452 movies match. French is the most common original language (112 movies), followed by Spanish (72), Italian (56) and German (51). The example title is the movie with most votes in each language, e.g. *Leon: The Professional* for French. xx is the code TMDB uses for "no language", here *Quest for Fire*, which has no dialogue, and it is kept as a value of original_language. Some codes look wrong in the data: A Civil Action (pt) and Killer Party (nl) are American films in English, so the smaller counts contain some noise.

```
================================================================================
Task 9: Among movies where original_language != "en" but at least one
production company or country is United States, which are the top 10 original
languages by count? For each language, report the count and one example title.
================================================================================
Matching movies: 452 (0 only through a company name with 'United States').

First result document:
{'_id': 'fr',
 'movies': 112,
 'example_title': 'Leon: The Professional',
 'example_vote_count': 4293}
```

rank	original_language	movies	example_title	example_vote_count
1	fr	112	Leon: The Professional	4,293
2	es	72	Pan's Labyrinth	3,122
3	it	56	The Good, the Bad and the Ugly	2,371
4	de	51	Bagdad Cafe	76
5	ja	31	Dragon Ball Z: Resurrection 'F'	377
6	pt	15	A Civil Action	113
7	xx	14	Quest for Fire	111
8	nl	12	Killer Party	12
9	ru	11	Brother 2	39
10	zh	11	Crouching Tiger, Hidden Dragon	949

```
The production companies have no country in the data, so 'company is United States' can only
be checked by name. TMDB uses the code 'xx' for movies with no language (e.g. silent films).
[task 9 took 0.1 s]
```

Output 17: Result of task 9.

Task 10: Rating count, variance and genre diversity per user

The ratings are grouped by user to get the number of ratings, the population standard deviation with `$stdDevPop` (squared to the variance) and the set of rated movie IDs. A `$lookup` on the movies collection then fetches the genres of these movies, and they are combined with `$setUnion`. Since the movie IDs are collected first, the `$lookup` runs once per user and not once per rating. Ratings that could not be linked to a movie count in the number and the variance, but contribute no genres. The result for all 270,896 users is written to the collection `user_rating_stats` with `$out` (and to `output/task10_user_rating_stats.csv`). The two top-10 lists are then simple `find()` queries with `sort` and `limit` on this collection, using the users with ≥ 20 ratings in both lists. The aggregation over the 26 million ratings takes about two minutes.

As in task 3, many users reach the maximum: 2,896 of the 166,444 users with ≥ 20 ratings have rated movies in all 20 genres, and we again break the tie by the fewest ratings (user 12420 covers all 20 genres with only 53 ratings). No user with fewer than 20 ratings has rated all 20 genres, so the limit does not change this list. The variance also has a maximum, 5.0625, reached when half of the ratings are 0.5 and the other half 5.0. 6 users with ≥ 20 ratings have exactly this variance, and we list those with the most ratings first. These users rate everything as either the worst or

the best possible, which is probably a different use of the scale (like/dislike) rather than strong opinions.

```
================================================================================
Task 10: For each user, what is their ratings count, population variance of
ratings, and number of distinct genres rated? List the top 10 most genre-
diverse users, and separately the top 10 highest-variance users (only users
with >= 20 ratings).
================================================================================
Statistics for 270,896 users written to the collection user_rating_stats (126 s).
Users with >= 20 ratings: 166,444, of these with all 20 genres: 2,896
Users with < 20 ratings and all 20 genres: 0
Many users are tied at the maximum, so the top 10 is decided by the tie-breaker.

First result document:
{'_id': 12420, 'ratings': 53, 'variance': 0.560342, 'distinct_genres': 20}

Top 10 most genre-diverse users (ties: fewer ratings first):
  rank  user_id      ratings    variance    distinct_genres
------  ---------  ---------  ----------  -----------------
     1  12420             53      0.5603                20
     2  55755             55      2.3425                20
     3  238289            61      1.1516                20
     4  257021            62      2.5323                20
     5  73720             66      0.6451                20
     6  201607            67      0.9143                20
     7  202270            67      1.0956                20
     8  173031            70      1.8335                20
     9  12427             72      0.8233                20
    10  161691            73      0.4544                20

Users with >= 20 ratings at the highest possible variance 5.0625 (half 0.5, half 5.0): 6
First result document:
{'_id': 167241, 'ratings': 34, 'variance': 5.0625, 'distinct_genres': 14}

Top 10 highest-variance users (ties: more ratings first):
  rank  user_id      ratings    variance    distinct_genres
------  ---------  ---------  ----------  -----------------
     1  167241            34      5.0625                14
     2  185889            26      5.0625                14
     3  18703             20      5.0625                13
     4  57842             20      5.0625                14
     5  83102             20      5.0625                15
     6  153882            20      5.0625                15
     7  15395             83      5.0618                14
     8  71009             27      5.0556                13
     9  179759            21      5.0510                13
    10  221535           120      5.0400                19
(all 270,896 rows written to output/task10_user_rating_stats.csv)
[task 10 took 127.4 s]
```

Output 18: Result of task 10.

3 Discussion

3.1 Choices that go beyond the assignment text

Unknown is not zero. Budget, revenue, runtime and vote average use 0 for "unknown". If we had kept the zeros, the median revenue would be 0 for most directors in task 1, and the vote averages in tasks 1, 2 and 4 would be pulled down by movies without votes. Storing `null` instead lets MongoDB's accumulators ignore the unknown values, and we report how many values the answers in tasks 1, 4 and 8 rest on.

Ambiguous tasks. Several questions can be read in more than one way, and we chose as follows:

- Task 1: "≥ 5 movies" is answered both literally (≥ 5 directed movies) and with ≥ 5 movies with known revenue, so that the median is based on at least five values (see the task). The two lists share only 5 directors.
- Tasks 3 and 10: the top-10 lists hit a natural maximum (20 genres, variance 5.0625), so we document the number of ties and the tie-break (fewest movies/ratings for breadth, most ratings for variance).
- Task 6: "unknown gender may be ignored" is applied after taking the top 5 by `order`, so the proportion is among the known genders of the five top-billed. Decades with only a few movies are included as asked, but their averages are not meaningful.
- Task 7: we search only for "noir", which covers "neo-noir" (see the task), and return 5 movies because only 5 match.
- Task 8: a director acting in their own film is not counted as a director-actor pair.
- Task 9: companies have no country, so "a production company is United States" is checked on the company name.
- Task 10: the limit of ≥ 20 ratings is used for both lists, and ratings without a linked movie still count in the number of ratings and the variance.

3.2 MongoDB compared to MySQL

Compared to MySQL in assignment 2, MongoDB let us store the nested data as it is. In MySQL, the same data would need about ten tables (movies, people, genres, companies, countries, languages, keywords, and cast, crew and other junction tables), and most of the tasks would need several joins. With embedded documents, nine of the ten tasks only read the movies collection. The other side of this is that questions that combine elements within the same document, such as actor pairs (task 2) and director-actor pairs (task 8), require unwinding the same array twice. That is the document version of a self-join, and it creates millions of intermediate documents.

MongoDB does not require a schema, but we found that we wanted the same guarantees as in MySQL: required fields, types and value ranges. The `$jsonSchema` validators gave us this, similar to `NOT NULL` and `CHECK` in assignment 2. There are no

foreign keys, so the link between ratings and movies is only checked by our insert program. The aggregation pipelines are easy to build step by step, because each stage can be tested on its own. On the other hand, expressions such as the median or the genre counts in task 3 become deeply nested and are harder to read than the equivalent SQL.

3.3 Pain points and what we learned

The first problem was the file format. The nested columns are Python literals and not JSON, and three rows are broken in a way that does not cause an error but silently shifts the columns, which created fake genres. As with `DAY_TYPE` in assignment 2, we learned to check values that look complete: 0 for an unknown budget, "No overview found." and gender code 0 are all missing values in disguise. Linking the ratings also required understanding three ID systems (TMDB, IMDb and MovieLens) and resolving them during the insert.

The queries taught us that a "top 10" question is often a tie-breaking question when the measure has a maximum, and that text search depends on how the text is split into words. Choosing the schema (embedding versus referencing) took the most thought. Once that was decided, most questions became a single pipeline over one collection, and the heaviest queries were the self-join of the cast in task 2 (6.5 million pairs) and the one that crossed the two collections (task 10).

4 Feedback

- The introduction to the tasks says that Part 1 is about analysing and cleaning "the Porto dataset" (from assignment 2).
- In task 9, production companies in the dataset have only a name and an ID, not a country, so "a production company is United States" can only be checked by name. A note about this would help.

5 AI use

We used Claude Code, an AI coding assistant, during the assignment, with the assignment text and provided files as context. We used it together to plan the EDA checks, the cleaning rules, the database design and the aggregation pipelines. Parts of our implementation are built with its suggestions as inspiration. In addition, Claude Code was used to improve code quality and to debug our local MongoDB setup in Docker. For the report, Claude gave feedback on language, formulations and mistakes. All design decisions, interpretations and results were reviewed by us, and all numbers in the report come from running our own code against our database.

Appendix A: EDA console output

Complete output of `part1_eda.py` (`output/part1_eda.txt`). The output of check 1 is referred to as A1, etc.

```
================================================================================
1. Files
================================================================================
File                     MB  CSV rows      Columns  Column names
-------------------  -----  ----------  ---------  ----------------------------------------
movies_metadata.csv   34.4  45,466             24  adult, belongs_to_collection, budget, genres,
                                                   homepage, id, imdb_id, original_language,
                                                   original_title, overview, popularity,
                                                   poster_path, production_companies,
                                                   production_countries, release_date, revenue,
                                                   runtime, spoken_languages, status, tagline,
                                                   title, video, vote_average, vote_count
credits.csv          189.9  45,476              3  cast, crew, id
keywords.csv           6.2  46,419              2  id, keywords
links.csv              1    45,843              3  movieId, imdbId, tmdbId
links_small.csv        0.2  9,125               3  movieId, imdbId, tmdbId
ratings.csv          709.6  26,024,289          4  userId, movieId, rating, timestamp
ratings_small.csv      2.4  100,004             4  userId, movieId, rating, timestamp
```

A1: Files

```
================================================================================
2. movies_metadata.csv - rows split by a line break in the overview
================================================================================
  Fields in CSV row  Rows
-------------------  ------
                 10  3
                 15  3
                 24  45,460

The 15-field rows read with the 24 column names (all columns are shifted):
id          adult                          genres                                              original_language
----------  -----------------------------  --------------------------------------------------  -------------------
1997-08-20  - Written by Ørnås             [{'name': 'Carousel Productions', 'id': 11176},                    104
2012-09-29  Rune Balot goes to a casino c  [{'name': 'Aniplex', 'id': 2883}, {'name': 'GoHa                    68
2014-01-01  Avalanche Sharks tells the st  [{'name': 'Odyssey Media', 'id': 17161}, {'name'                    82

After joining each 10-field row with the 15-field row that follows it:
    id  title                              release_date    genres
------  ---------------------------------  --------------  ---------------------------------------
 82663  Midnight Man                       1997-08-20      Action, Thriller, Drama
122662  Mardock Scramble: The Third Exhaust  2012-09-29      Animation, Science Fiction
249260  Avalanche Sharks                   2014-01-01      TV Movie, Action, Horror, Science Fiction
-> 3 movies repaired, 45,463 movie rows in total.
```

A2: movies_metadata.csv - rows split by a line break in the overview

```
================================================================================
3. movies_metadata.csv - duplicate ids
================================================================================
-----------------------------------  ------
Movie rows                           45,463
Unique ids                           45,433
Ids that occur more than once        29
Extra (duplicate) rows               30
Duplicate groups that are exact copies  16
Groups where popularity differs      13
Groups where vote_count differs      2
-----------------------------------  ------

Examples of copies that differ (kept = highest vote_count, the first one on ties):
id      title                        vote_count    popularity    Decision
------  ---------------------------  ------------  ------------  ----------
132641  Wife                         0             0.096079      kept
132641  Wife                         0             0.619388      dropped
22649   A Farewell to Arms           29            1.914697      kept
22649   A Farewell to Arms           29            2.411191      dropped
84198   A Place at the Table         7             0.501046      kept
84198   A Place at the Table         7             1.673307      dropped
10991   Pokémon: Spell of the Unknown  143           10.264597     dropped
10991   Pokémon: Spell of the Unknown  144           6.480376      kept
-> 45,433 unique movies are inserted.
```

A3: movies_metadata.csv - duplicate ids

```
================================================================================
4. movies_metadata.csv - missing values per column (after repair and deduplication)
================================================================================
Column                Empty    0 / [] / placeholder    Stored as None or []    Share
--------------------  -------  ----------------------  ----------------------  -------
adult                 0        0                       0                       0.0 %
belongs_to_collection 40,945   0                       40,945                  90.1 %
budget                0        36,553                  36,797                  81.0 %
genres                0        2,442                   2,442                   5.4 %
homepage              37,659   0                       37,659                  82.9 %
id                    0        0                       0                       0.0 %
imdb_id               17       0                       17                      0.0 %
original_language     11       0                       11                      0.0 %
original_title        0        0                       0                       0.0 %
overview              959      142                     1,101                   2.4 %
popularity            0        0                       0                       0.0 %
poster_path           383      0                       -
production_companies  0        11,868                  11,868                  26.1 %
production_countries  0        6,279                   6,279                   13.8 %
release_date          84       0                       84                      0.2 %
revenue               0        38,035                  38,189                  84.1 %
runtime               257      1,558                   1,815                   4.0 %
spoken_languages      0        3,827                   3,827                   8.4 %
status                81       0                       81                      0.2 %
tagline               25,032   0                       25,032                  55.1 %
title                 0        0                       0                       0.0 %
video                 0        0                       0                       0.0 %
vote_average          0        2,995                   2,995                   6.6 %
vote_count            0        2,896                   0 kept
budget/revenue: also None when below 1000 USD. vote_average: None when 0 (no votes or 0 with votes).
poster_path is not stored. Placeholders: No Overview, No overview found., No overview yet.
```

A4: movies_metadata.csv - missing values per column (after repair and deduplication)

```
================================================================================
5. movies_metadata.csv - numeric attributes (known values after cleaning)
================================================================================
Attribute     Known    min    p25         median       p75          max             mean
-----------  -------  -----  ---------  ----------  ----------  -------------  ----------
budget        8,636    1,000  2,200,000  9,000,000   26,000,000  380,000,000    22,224,860
revenue       7,244    1,081  2,955,039  17,693,675  69,411,370  2,787,965,087  70,320,445
runtime       43,618   1      86         95          107         1,256          97
vote_average  42,438   0.5    5.3        6.1         6.9         10.0           6.0
vote_count    45,433   0      3          10          34          14,075         110
popularity    45,433   0.00   0.39       1.13        3.68        547.49         2.92
budget/revenue in USD, runtime in minutes. vote_count includes the movies with 0 votes.
```

A5: movies_metadata.csv - numeric attributes (known values after cleaning)

```
================================================================================
6. movies_metadata.csv - suspicious values
================================================================================
-----------------------------------  -----
Budget > 0 but below 1,000 USD         244
Revenue > 0 but below 1,000 USD        154
Budget 1,000 to 9,999 USD (kept)        54
Revenue 1,000 to 9,999 USD (kept)       83
vote_average 0 although vote_count > 0  99
vote_average 0 with only 1 vote         81
Lowest vote_average above 0              0.5
Runtime above 6 hours (kept)            67
Movies marked adult                      9
-----------------------------------  -----

Tiny budgets/revenues, the most voted movies (set to None, i.e. unknown):
title                         year    budget    revenue    vote_count
----------------------------  ------  --------  ---------  ------------
Modern Times                  1936    1         8500000    881
Ong-Bak: The Thai Warrior     2003    0         15         556
Irreversible                  2002    0         792        550
Death at a Funeral            2007    9000000   46         508
The Karate Kid, Part II       1986    113       115103979  457
I Spit on Your Grave 2        2013    0         809        281
2:22                          2017    0         422        277
Dr. Horrible's Sing-Along Blog  2008    200000    3          236

Longest runtimes (mostly TV mini-series, kept):
title                          year    runtime (min)
-----------------------------  ------  ---------------
Centennial                     1978             1256
Jazz                           2001             1140
Baseball                       1994             1140
Berlin Alexanderplatz          1980              931
Heimat: A Chronicle of Germany  1984              925
The Story of Film: An Odyssey  2011              900
```

A6: movies_metadata.csv - suspicious values

```
================================================================================
7. movies_metadata.csv - categorical attributes
================================================================================
status           Movies    Share
---------------  --------  -------
Released         44,988    99.02 %
Rumored          229       0.50 %
Post Production  98        0.22 %
(missing)        81        0.18 %
In Production    20        0.04 %
Planned          15        0.03 %
Canceled         2         0.00 %

original_language: 89 distinct codes. The 10 most common:
original_language    Movies    Share
-------------------  --------  -------
en                   32,251    71.0 %
fr                   2,436     5.4 %
it                   1,529     3.4 %
ja                   1,347     3.0 %
de                   1,079     2.4 %
es                   994       2.2 %
ru                   826       1.8 %
hi                   508       1.1 %
ko                   444       1.0 %
zh                   409       0.9 %
Missing original_language: 11

Flag      Movies with True
------  ------------------
adult                    9
video                   93
```

A7: movies_metadata.csv - categorical attributes

```
================================================================================
8. movies_metadata.csv - release_date
================================================================================
-------------------------------------------------  ----------
Movies with release_date                           45,349
Missing release_date                               84
Earliest                                           1874-12-09
Latest                                             2020-12-16
Release date after the last rating (2017-08-04)    31
Release date after the last rating, status Released  17
-------------------------------------------------  ----------

Decade    Movies
--------  --------
1870s     2
1880s     4
1890s     75
1900s     87
1910s     176
1920s     431
1930s     1,315
1940s     1,493
1950s     2,079
1960s     2,621
1970s     3,471
1980s     3,928
1990s     5,676
2000s     11,195
2010s     12,795
2020s     1
```

A8: movies_metadata.csv - release_date

```
================================================================================
9. movies_metadata.csv - genres
================================================================================
20 distinct genres (after repairing the split rows):
Genre            Movies    Share    As first genre
---------------  --------  -------  ----------------
Drama            20,244    44.6 %   11,953
Comedy           13,176    29.0 %   8,816
Thriller         7,619     16.8 %   1,663
Romance          6,730     14.8 %   1,191
Action           6,592     14.5 %   4,487
Horror           4,671     10.3 %   2,619
Crime            4,304     9.5 %    1,684
Documentary      3,930     8.7 %    3,413
Adventure        3,490     7.7 %    1,509
Science Fiction  3,044     6.7 %    647
Family           2,767     6.1 %    524
Mystery          2,464     5.4 %    554
Fantasy          2,309     5.1 %    703
Animation        1,931     4.3 %    1,124
Foreign          1,619     3.6 %    118
Music            1,597     3.5 %    487
History          1,398     3.1 %    279
War              1,322     2.9 %    379
Western          1,042     2.3 %    451
TV Movie         766       1.7 %    390

Genres per movie    Movies
------------------  --------
0                   2,442
1                   14,552
2                   14,471
3                   9,578
4                   3,377
5+                  1,013
```

A9: movies_metadata.csv - genres

```
================================================================================
10. movies_metadata.csv - belongs_to_collection
================================================================================
-----------------------------------  -----
Movies in a collection               4,488
Distinct collections                 1,695
Collections with >= 3 movies         628
Collection ids with more than one name  0
-----------------------------------  -----

Movies in collection      Collections
----------------------  -------------
1                                390
2                                677
3-5                              526
6-10                              83
> 10                              19

Largest collections:
Collection                      Movies
----------------------------  --------
The Bowery Boys                     29
Totò Collection                     27
James Bond Collection               26
Zatôichi: The Blind Swordsman       26
The Carry On Collection             25
```

A10: movies_metadata.csv - belongs_to_collection

```
================================================================================
11. movies_metadata.csv - production companies, countries and spoken languages
================================================================================
Column               Distinct    Empty list    Share    Most common
-------------------  ----------  ------------  -------  ----------------------------------------------
production_companies  23,538      11,868        26.1 %   Warner Bros. (1,250), Metro-Goldwyn-Mayer (MGM)
                                                          (1,074), Paramount Pictures (1,001)
production_countries  161         6,279         13.8 %   US (21,142), GB (4,092), FR (3,932)
spoken_languages      133         3,827         8.4 %    en (28,729), fr (4,193), de (2,624)

-----------------------------------------------  ------------------------------------------------
Movies with United States as production country  21,142
United States and original_language not 'en'     452
Company names containing 'United States'         United States Pictures (7), United States Navy (1)
-----------------------------------------------  ------------------------------------------------
-> Companies have only id and name, no country.
```

A11: movies_metadata.csv - production companies, countries and spoken languages

```
================================================================================
12. credits.csv - rows and duplicates
================================================================================
--------------------------------------------------------  ------
Rows                                                      45,476
Unique movie ids                                          45,432
Ids that occur more than once                             43
Extra (duplicate) rows                                    44
Duplicate groups that are exact copies                    36
Duplicate groups with the same cast                       43
Duplicate groups with the same crew members (credit_id)   43
Credits for movie ids not in movies_metadata              0
Movies without a credits row                              1
--------------------------------------------------------  ------
```

A12: credits.csv - rows and duplicates

```
================================================================================
13. credits.csv - cast and crew (after deduplication)
================================================================================
-----------------------------------  ------------------
Movies with empty cast               2,415
Movies with empty crew               772
Cast entries / median / max per movie  562,044 / 10 / 313
Crew entries / median / max per movie  463,836 / 6 / 435
-----------------------------------  ------------------

Gender (TMDB code 0 = not set, stored as None):
                 Entries    female    male    unknown
---------------  ---------  --------  ------  ---------
All cast entries  562,044    19.9 %    40.3 %  39.8 %
Top-5 billed cast  202,303    25.5 %    46.0 %  28.5 %
Crew entries      463,836    6.7 %     34.6 %  58.6 %

Cast order (movies with a non-empty cast: 43,018):
-----------------------------------------  -------------------
Cast order consecutive 0..n-1              31,576
Cast order gaps or other start             11,355
Cast order duplicate order values          87
Movies where an actor has more than one role  951
Movies without a Director in the crew      888
Movies with 1 / 2 / 3+ directors           41,350 / 2,714 / 481
Max. directors of one movie                42
Movies where a director is also in the cast  3,754
-----------------------------------------  -------------------

-----------------------------------  ------------------------------------------------------------------------
Distinct actors (person ids)         206,158
Distinct actor names                 202,747
Names used by more than one person id  3,053
Examples                             Varvara Dvalishvili (7 ids), Colin Campbell (6 ids), Sam Harris (6 ids)
Distinct actor pairs per movie, summed  6,487,453
-----------------------------------  ------------------------------------------------------------------------
-> People are identified by their TMDB person id, the name is only used for display.
```

A13: credits.csv - cast and crew (after deduplication)

```
======================================================================
14. keywords.csv
======================================================================
-------------------------------------------  -------
Rows                                         46,419
Unique movie ids                             45,432
Extra (duplicate) rows                       987
Duplicate groups that are not exact copies   0
Keyword rows for ids not in movies_metadata  0
Movies without keywords (empty list or no row)  14,341
Keywords per movie: median / max             2 / 149
Distinct keywords                            19,956
Movies with keyword 'film noir' or 'neo-noir'   398
Noir keyword and 'noir' in overview or tagline  18
-------------------------------------------  -------

Most common keywords: woman director (3,039), independent film (1,914), murder (1,285), based on novel (822), musical (726), sex (679), violence (647), nudity (629)
```

A14: keywords.csv

```
================================================================================
15. links.csv / links_small.csv
================================================================================
File            MovieLens ids      Empty tmdbId    tmdbId not in movies    tmdbIds with > 1 MovieLens id
--------------  ---------------  --------------  ----------------------  ------------------------------
links.csv       45,843                      219                     161                              29
links_small.csv  9,125                       13                      30                               0
MovieLens ids in links_small.csv but not in links.csv: 10. Ids in both with a different tmdbId: 0.
```

A15: links.csv / links_small.csv

```
================================================================================
16. ratings.csv (the full file with 26 million ratings)
================================================================================
-----------------------------------------  ---------------------------------------
Ratings                                    26,024,289
Users                                      270,896
Rated MovieLens movies                     45,115
First / last rating (UTC)                  1995-01-09 11:46:44 / 2017-08-04 06:57:50
Ratings per user: min / median / mean / max  1 / 30 / 96.1 / 18,276
Users with >= 20 ratings                   166,444
Users with 1 rating                        5,084
Ratings per movie: median / max            8 / 91,921
-----------------------------------------  ---------------------------------------

  Rating  Ratings    Share
--------  ---------  -------
     0.5  404,897    1.6 %
     1    843,310    3.2 %
     1.5  403,607    1.6 %
     2    1,762,440  6.8 %
     2.5  1,255,358  4.8 %
     3    5,256,722  20.2 %
     3.5  3,116,213  12.0 %
     4    6,998,802  26.9 %
     4.5  2,170,441  8.3 %
     5    3,812,499  14.6 %

Link to movies                                          Ratings     Share
------------------------------------------------------  ----------  -------
Linked to a movie                                       25,981,582  99.84 %
tmdbId not in movies_metadata                           29,204      0.11 %
Movie has no tmdbId                                     13,503      0.05 %
(Linked) same user and TMDB movie as an earlier rating  125         0.00 %
(The data is sorted by user, so the same user rating one TMDB movie twice is found per user.)

ratings_small.csv for comparison: 100,004 ratings, 671 users, 9,066 movies (not used).
```

A16: ratings.csv (the full file with 26 million ratings)

```
================================================================================
17. Impact of the missing values on the Part 2 tasks
================================================================================
Group          Movies    budget known    revenue known    runtime known    vote_average known
-------------  --------  --------------  ---------------  ---------------  --------------------
1870s          2         0.0 %           0.0 %            100.0 %          100.0 %
1880s          4         0.0 %           0.0 %            100.0 %          100.0 %
1890s          75        0.0 %           0.0 %            100.0 %          100.0 %
1900s          87        2.3 %           0.0 %            100.0 %          90.8 %
1910s          176       4.0 %           2.3 %            99.4 %           89.8 %
1920s          431       9.0 %           3.7 %            98.1 %           89.1 %
1930s          1,315     6.0 %           2.9 %            99.0 %           84.9 %
1940s          1,493     5.2 %           3.4 %            98.9 %           91.4 %
1950s          2,079     6.8 %           4.5 %            97.7 %           93.2 %
1960s          2,621     8.6 %           6.1 %            97.3 %           92.5 %
1970s          3,471     10.4 %          7.3 %            96.5 %           90.5 %
1980s          3,928     18.7 %          22.5 %           96.7 %           94.5 %
1990s          5,676     22.5 %          22.9 %           96.1 %           93.2 %
2000s          11,195    25.2 %          19.7 %           95.5 %           94.4 %
2010s          12,795    22.4 %          17.5 %           95.0 %           94.9 %
2020s          1         0.0 %           0.0 %            0.0 %            0.0 %
US production  21,142    29.3 %          26.2 %           99.2 %           95.4 %
Not US         24,291    10.1 %          7.0 %            93.2 %           91.7 %

                         Movies    US production    Released 1990 or later
-----------------------  --------  ---------------  ----------------------
Movies with known revenue  7,244     77 %             79 %
All movies                 45,433    47 %             65 %
-> Missing budget/revenue is not spread evenly over decades and countries (compare the rows),
   so medians and totals of revenue (tasks 1, 4 and 8) mostly describe the movies where it is known.

-----------------------------------------------------  -------------
Task 1: directors with >= 5 movies                     2,281
Task 1: ... of these with >= 5 known revenues          382
Task 4: collections with >= 3 movies                   628
Task 4: ... with all revenues known                    129
Task 4: ... with no revenue known                      274
Task 5: movies with date and genre but unknown runtime  1,271
Task 7/8: movies with vote_count >= 50 / >= 100        9,142 / 6,050
-----------------------------------------------------  -------------

Task 6: share of the top-5 billed cast with unknown gender, per decade:
Decade    Top-5 cast entries    Unknown gender
--------  --------------------  ---------------
1870s     2                     50.0 %
1880s     1                     100.0 %
1890s     68                    97.1 %
1900s     96                    77.1 %
1910s     775                   45.0 %
1920s     1,928                 38.6 %
1930s     6,296                 24.5 %
1940s     7,153                 17.6 %
1950s     9,943                 20.6 %
1960s     12,265                29.0 %
1970s     16,079                31.2 %
1980s     18,118                28.1 %
1990s     25,876                21.7 %
2000s     48,659                27.7 %
2010s     54,975                34.0 %
2020s     5                     0.0 %
-> Unknown gender is ignored in task 6. Where the share is high, the female proportion is
   based on fewer people per movie.

Figures written to report/figures/
```

A17: Impact of the missing values on the Part 2 tasks

Appendix B: Complete output of task 5

decade	primary_genre	movies	movies_with_runtime	median_runtime
1870s	Documentary	2	2	1.00
1880s	Documentary	4	4	1.00
1890s	Fantasy	8	8	1.50
1890s	Documentary	27	27	1.00
1890s	Comedy	9	9	1.00
1890s	Horror	3	3	1.00
1890s	Action	2	2	1.00
1890s	Drama	2	2	1.00
1890s	Family	1	1	1.00
1890s	History	1	1	1.00
1900s	History	1	1	60.00
1900s	Adventure	4	4	15.00
1900s	Drama	7	7	10.00
1900s	Action	2	2	9.00
1900s	Thriller	1	1	8.00
1900s	Science Fiction	2	2	5.50
1900s	Fantasy	15	15	4.00
1900s	Horror	2	2	4.00
1900s	Comedy	17	17	3.00
1900s	Music	1	1	3.00
1900s	Crime	2	2	2.00
1900s	Animation	1	1	2.00
1900s	Documentary	12	12	1.00
1900s	Romance	2	2	1.00
1910s	History	2	2	109.00
1910s	War	2	2	108.00
1910s	Mystery	3	3	84.00
1910s	Adventure	8	8	70.00
1910s	Romance	6	6	68.50
1910s	Drama	52	52	65.00
1910s	Western	4	4	60.50
1910s	Crime	7	7	59.00
1910s	Family	1	1	52.00
1910s	Documentary	2	2	43.50
1910s	Horror	2	2	38.50
1910s	Action	1	1	29.00
1910s	Comedy	54	53	27.00
1910s	Fantasy	6	6	14.50
1910s	Animation	5	5	12.00
1920s	Science Fiction	1	1	120.00
1920s	Music	2	2	114.00
1920s	War	4	4	106.00
1920s	History	4	4	105.50
1920s	Adventure	22	21	100.00
1920s	Thriller	4	4	93.50
1920s	Action	18	18	90.50
1920s	Drama	165	164	90.00
1920s	Western	10	10	88.50
1920s	Romance	18	18	88.00
1920s	Crime	16	16	83.00
1920s	Horror	5	5	83.00
1920s	Comedy	93	92	67.00
1920s	Documentary	18	18	67.00
1920s	Fantasy	4	4	66.00
1920s	Mystery	2	2	57.50
1920s	Family	2	2	37.00
1920s	Animation	10	10	6.50
1930s	Music	32	32	94.50
1930s	History	10	10	91.00
1930s	Adventure	55	55	90.00

Task 5, part 1 of 4.

decade	primary_genre	movies	movies_with_runtime	median_runtime
1930s	Romance	80	79	86.00
1930s	Drama	382	382	85.00
1930s	War	13	13	85.00
1930s	Comedy	353	349	82.00
1930s	Western	22	21	81.00
1930s	Science Fiction	5	5	80.00
1930s	Thriller	10	10	79.00
1930s	Family	5	5	78.00
1930s	Fantasy	8	8	76.50
1930s	Action	80	80	76.00
1930s	Crime	92	92	74.00
1930s	Mystery	38	38	70.00
1930s	Horror	28	28	68.00
1930s	Documentary	14	14	63.00
1930s	Animation	37	37	8.00
1940s	Science Fiction	1	1	220.00
1940s	History	11	11	107.00
1940s	Adventure	45	44	98.50
1940s	Drama	468	466	98.00
1940s	Action	88	88	93.50
1940s	Music	35	35	92.00
1940s	Romance	52	52	91.50
1940s	War	41	41	91.00
1940s	Comedy	326	323	90.00
1940s	Crime	136	135	86.00
1940s	Thriller	47	47	86.00
1940s	Western	41	40	86.00
1940s	Family	8	8	86.00
1940s	Fantasy	23	23	85.00
1940s	Mystery	41	41	71.00
1940s	Horror	27	27	67.00
1940s	Documentary	20	20	55.00
1940s	Animation	34	34	8.00
1950s	Music	34	34	101.00
1950s	Drama	636	627	98.00
1950s	History	17	17	97.00
1950s	War	46	46	96.00
1950s	Romance	77	76	95.00
1950s	Comedy	350	330	93.00
1950s	Action	167	166	93.00
1950s	Adventure	100	100	93.00
1950s	Mystery	22	22	91.50
1950s	Family	13	12	88.50
1950s	Thriller	46	46	87.50
1950s	Crime	170	170	86.00
1950s	Western	110	109	84.00
1950s	Fantasy	21	21	79.00
1950s	Horror	79	79	78.00
1950s	Science Fiction	50	50	78.00
1950s	Documentary	32	32	33.00
1950s	Animation	27	27	7.00
1960s	War	52	50	112.50
1960s	TV Movie	6	5	112.00
1960s	History	25	24	110.00
1960s	Drama	724	714	102.00
1960s	Romance	76	75	102.00
1960s	Mystery	36	36	101.50
1960s	Crime	107	104	100.50
1960s	Action	252	247	100.00
1960s	Music	29	28	100.00

Task 5, part 2 of 4.

decade	primary_genre	movies	movies_with_runtime	median_runtime
1960s	Western	99	96	98.50
1960s	Comedy	497	480	98.00
1960s	Adventure	139	138	97.00
1960s	Thriller	55	54	96.50
1960s	Foreign	10	10	91.50
1960s	Fantasy	30	30	91.00
1960s	Horror	146	146	87.00
1960s	Science Fiction	58	57	85.00
1960s	Documentary	77	77	80.00
1960s	Family	20	20	80.00
1960s	Animation	36	36	25.00
1970s	History	23	22	120.00
1970s	War	36	35	107.00
1970s	Romance	61	59	103.00
1970s	Drama	873	861	102.00
1970s	Crime	153	150	100.00
1970s	Mystery	57	56	99.50
1970s	Action	468	455	98.00
1970s	Thriller	135	126	98.00
1970s	Adventure	124	123	98.00
1970s	Music	33	33	98.00
1970s	Foreign	20	20	96.50
1970s	Comedy	512	492	96.00
1970s	Western	89	88	96.00
1970s	Science Fiction	49	47	94.00
1970s	Fantasy	48	47	94.00
1970s	Horror	321	320	91.00
1970s	Family	41	39	90.00
1970s	TV Movie	28	28	87.50
1970s	Documentary	139	136	86.00
1970s	Animation	60	57	38.00
1980s	War	32	31	120.00
1980s	Drama	858	840	105.00
1980s	Crime	136	132	102.00
1980s	TV Movie	29	29	102.00
1980s	Adventure	169	168	101.00
1980s	History	17	17	100.00
1980s	Action	494	493	98.00
1980s	Thriller	87	85	98.00
1980s	Romance	71	68	98.00
1980s	Mystery	39	38	97.00
1980s	Western	10	10	97.00
1980s	Comedy	860	823	96.00
1980s	Science Fiction	83	81	95.00
1980s	Music	39	39	94.00
1980s	Fantasy	91	90	93.50
1980s	Horror	382	379	91.00
1980s	Family	43	42	90.00
1980s	Foreign	7	7	89.00
1980s	Documentary	128	124	84.00
1980s	Animation	117	113	70.00
1990s	War	24	23	115.00
1990s	History	34	33	111.00
1990s	Drama	1,460	1,427	105.00
1990s	Crime	183	179	105.00
1990s	Western	22	21	102.00
1990s	Foreign	17	16	100.50
1990s	Mystery	57	56	100.00
1990s	Thriller	170	163	99.00
1990s	Romance	146	141	99.00

Task 5, part 3 of 4.

decade	primary_genre	movies	movies_with_runtime	median_runtime
1990s	Action	698	690	98.00
1990s	Adventure	198	193	97.00
1990s	Comedy	1,189	1,153	96.00
1990s	Fantasy	136	132	96.00
1990s	TV Movie	73	70	95.00
1990s	Science Fiction	97	94	94.00
1990s	Music	60	58	94.00
1990s	Horror	237	230	93.00
1990s	Family	75	75	90.00
1990s	Documentary	256	246	86.50
1990s	Animation	142	140	74.00
2000s	War	54	54	114.00
2000s	History	49	46	110.50
2000s	Drama	3,135	3,028	102.00
2000s	Crime	362	351	102.00
2000s	Romance	210	201	102.00
2000s	Adventure	340	335	100.00
2000s	Action	1,166	1,135	98.00
2000s	Thriller	330	313	98.00
2000s	Fantasy	170	167	98.00
2000s	Music	101	93	98.00
2000s	Foreign	36	33	98.00
2000s	Western	16	16	98.00
2000s	Mystery	100	99	96.00
2000s	Comedy	2,232	2,157	95.00
2000s	Science Fiction	115	111	92.00
2000s	Horror	603	590	91.00
2000s	Family	112	105	90.00
2000s	TV Movie	104	100	90.00
2000s	Documentary	1,044	995	88.00
2000s	Animation	326	322	76.00
2010s	History	84	81	105.00
2010s	Crime	319	309	103.00
2010s	Drama	3,186	3,082	100.00
2010s	Action	1,048	1,025	100.00
2010s	War	75	72	100.00
2010s	Romance	392	368	99.00
2010s	Music	121	117	99.00
2010s	Western	28	28	98.50
2010s	Adventure	304	297	97.00
2010s	Foreign	28	26	96.50
2010s	Mystery	157	152	95.50
2010s	Thriller	778	744	95.00
2010s	Comedy	2,323	2,143	93.00
2010s	Fantasy	143	137	93.00
2010s	Science Fiction	185	183	91.00
2010s	Horror	784	762	90.00
2010s	Family	203	189	90.00
2010s	Documentary	1,627	1,562	88.50
2010s	TV Movie	150	138	86.50
2010s	Animation	328	323	80.00
2020s	Action	1	0	

Task 5, part 4 of 4.